Pipelined ADC Implementation

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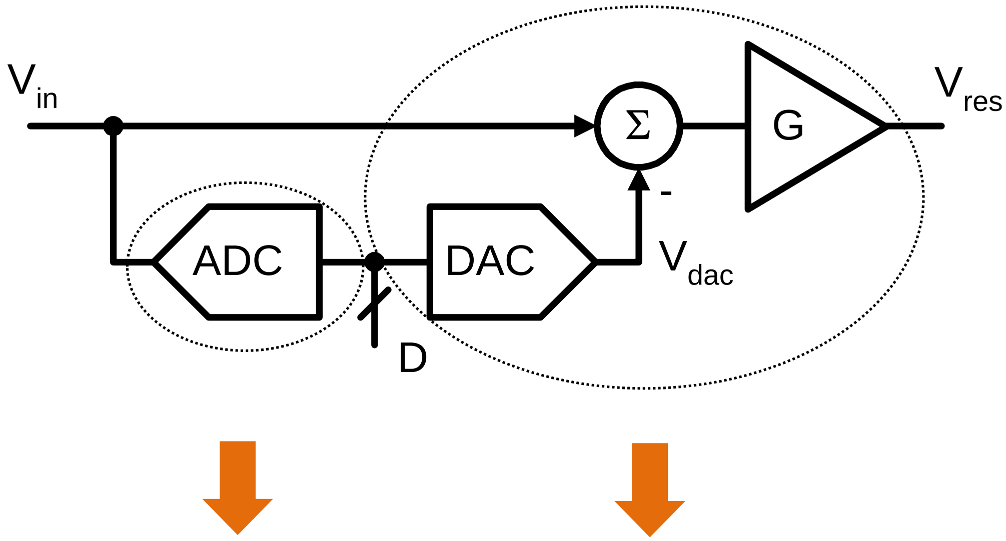
Stage Implementation



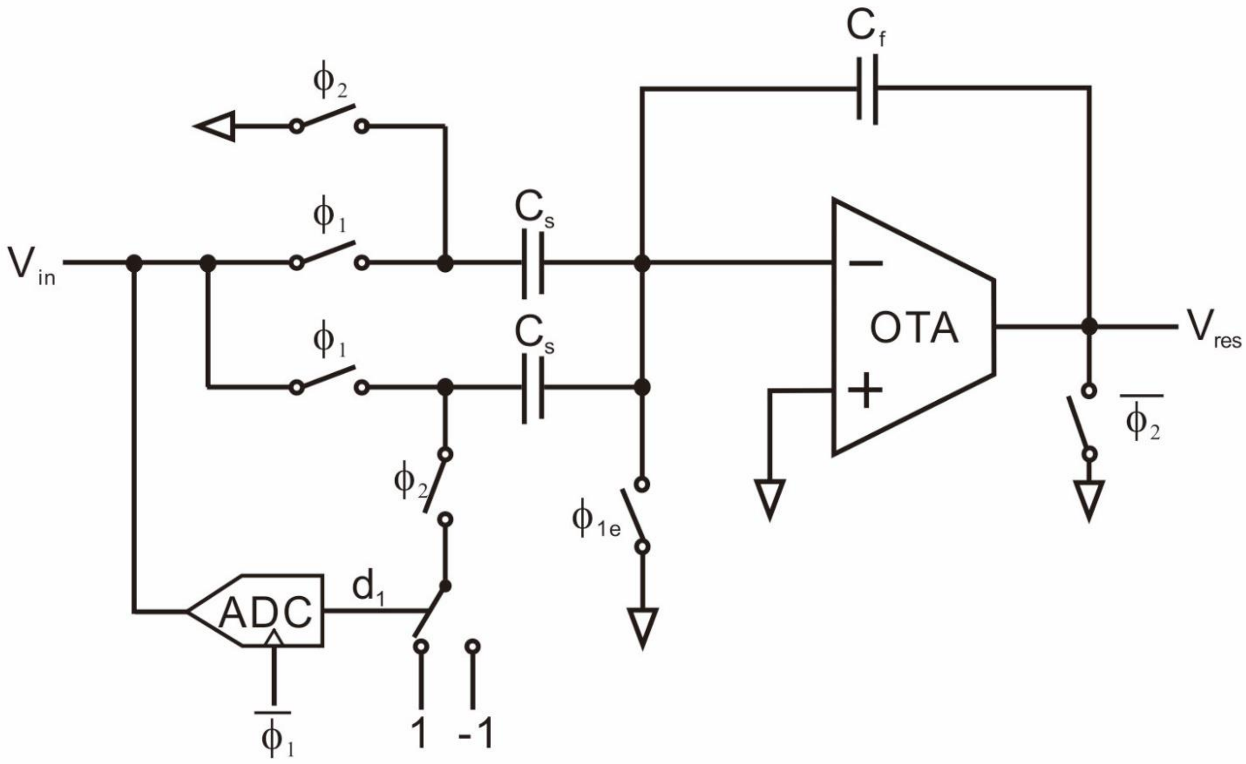
Flash ADC

“Multiplying DAC"

Switched-Capacitor Circuit

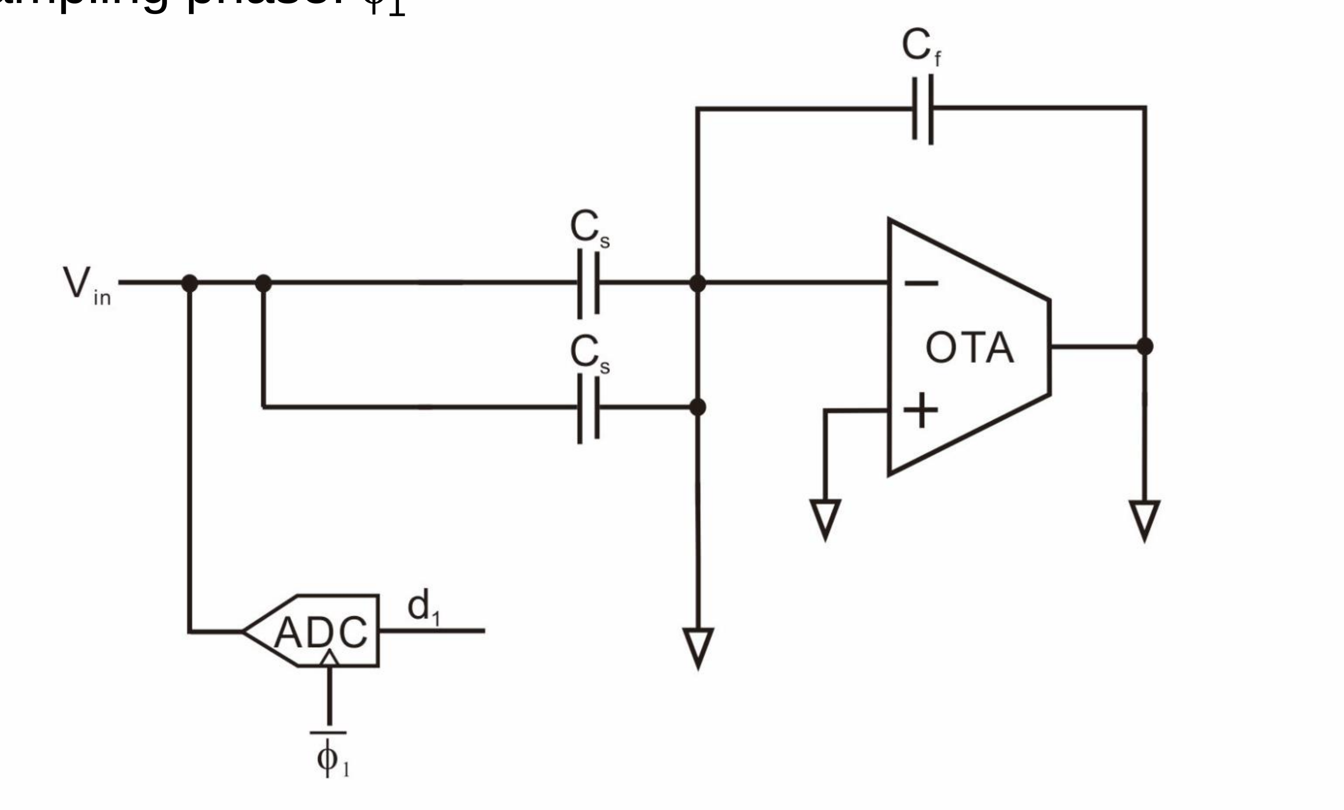
1-Bit Example (1)

- Switched capacitor implementation



1-Bit Example (2)

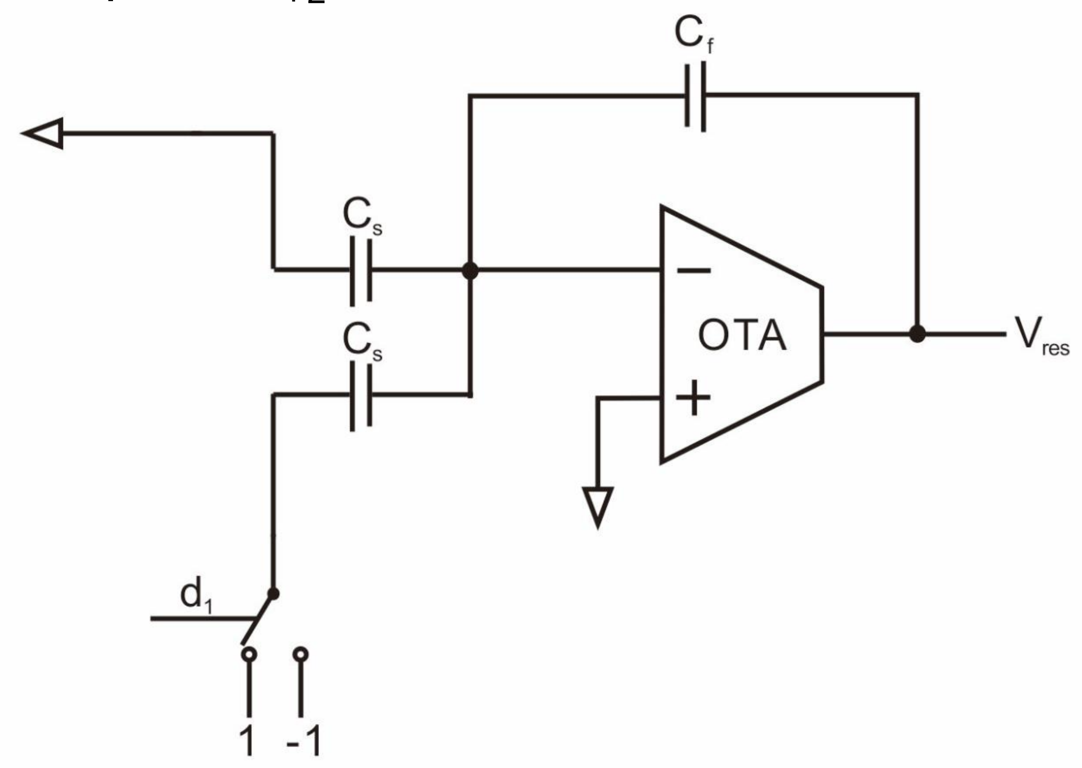
- Sampling phase: ϕ_1



$$Q_1 = -2C_s V_{in}$$

1-Bit Example (3)

- Charge transfer phase: ϕ_2



$$Q_2 = -C_s d_1 - C_f V_{res}$$

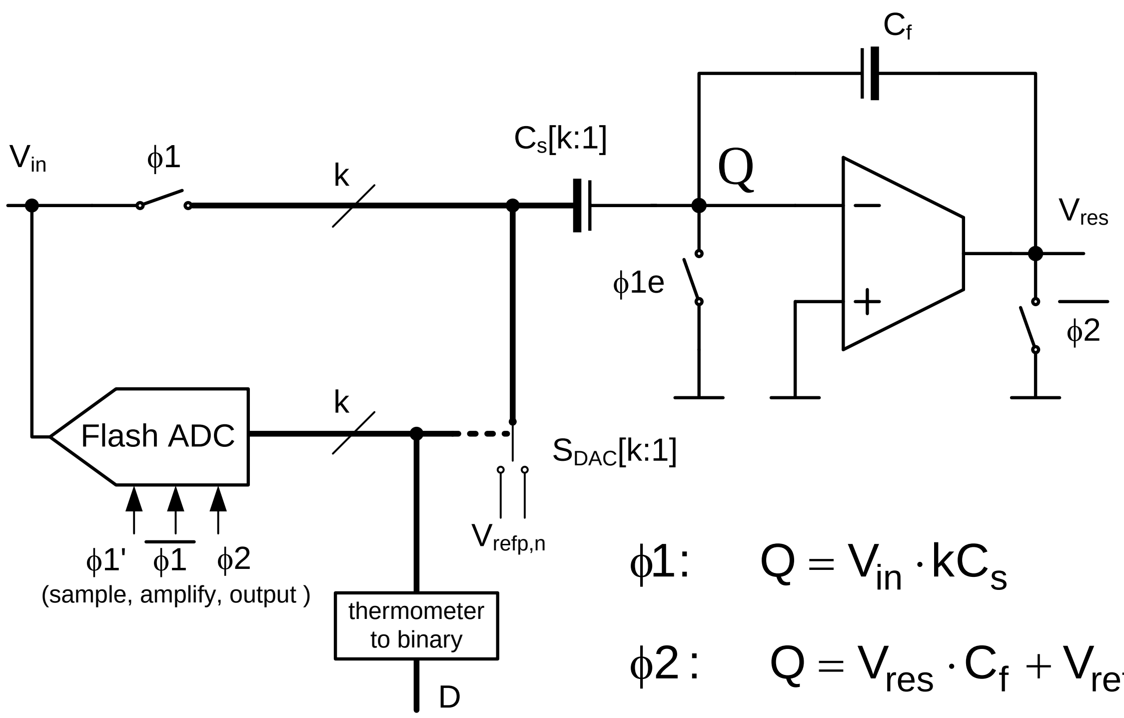
1-Bit Example (4)

- Due to charge conservation:

$$-2C_s V_{in} = Q_1 = Q_2 = -C_s d_1 - C_f V_{res}$$

$$\begin{aligned} V_{res} &= \frac{2C_s}{C_f}\left(V_{in} - \frac{d_1}{2}\right) \\ &= 2(V_{in} - V_{DAC}) \end{aligned}$$

Generic Circuit



$$\phi 1: \quad Q = V_{in} \cdot kC_s$$

$$\phi 2: \quad Q = V_{res} \cdot C_f + V_{refp} \cdot mC_s + V_{refn} \cdot (k-m)C_s$$

$$\therefore -V_{res} = \frac{kC_s}{C_f} V_{in} - \frac{mC_s}{C_f} V_{refp} + \frac{(k-m)C_s}{C_f} V_{refn}$$

$$= G \cdot (V_{in} - V_{dac})$$

List of Design Parameters

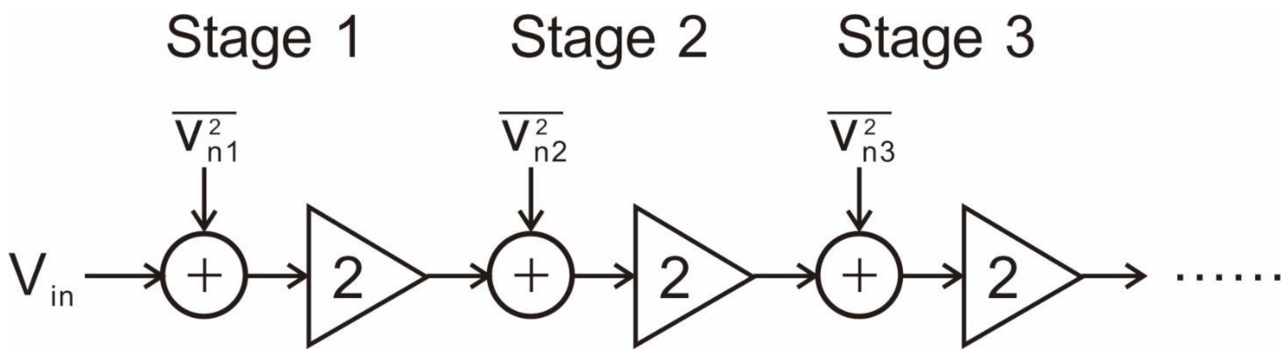
- Stage resolution, stage scaling factor
- Stage redundancy
- Thermal noise to quantization noise ratio
- OTA architecture
  - OTA sharing
- Switch topologies
- Comparator architecture
- Front-end SHA vs. SHA-less design
- Calibration approach (if needed)
- Time interleaving
- Technology and technology options (e.g. capacitors)

Thermal Noise Considerations

- Total input referred noise
  - Thermal noise + quantization noise
  - Costly to make thermal noise smaller than quantization noise
- Example: V_{FS}=1V, 10-bit ADC
  - $N_{quant}=LSB^2/12=(1V/2^{10})^2/12=(280\mu Vrms)^2$
  - Design for total input referred thermal noise ~280μVrms or larger, if SNR target allows
- Total input referred thermal noise
  - The sum of thermal noise from all stages
  - How should we distribute the total thermal noise among stages?

Stage Scaling (1)

- Example: Pipeline using 1-bit (effective) stages (G=2)



Noise

$$\overline{V_n{}^2} \propto \frac{kT}{C}$$

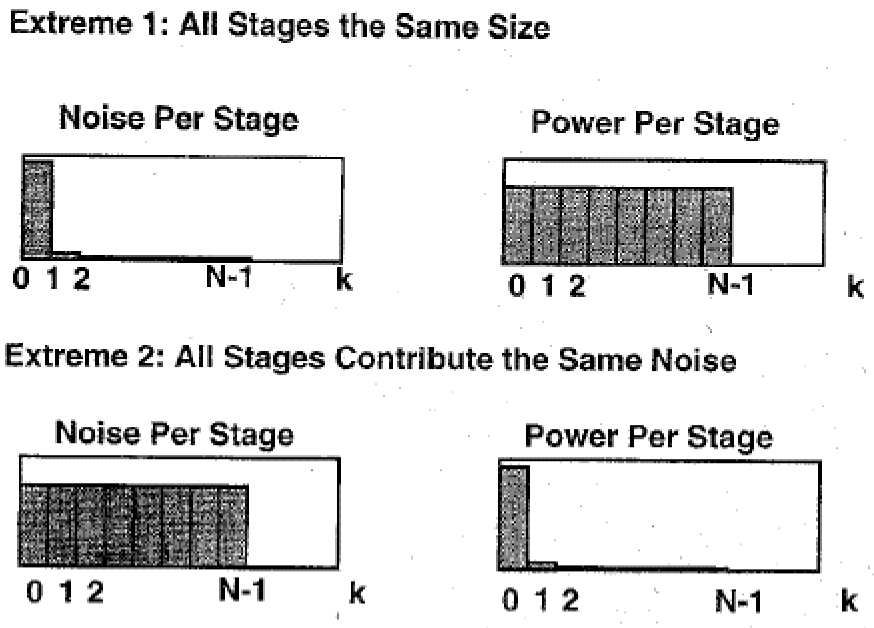
$$N_{tot} \propto kT\left[\frac{1}{C_1} + \frac{1}{4C_2} + \frac{1}{16C_3} + \cdots\right]$$

Power

$$P \propto C$$

$$P_{tot} \propto [C_1 + C_2 + C_3 + \cdots]$$

Stage Scaling (2)



[Cline 1996]

- Optimal capacitior scaling lies in between these two extremes

Stage Scaling (3)

- We want to minimize:

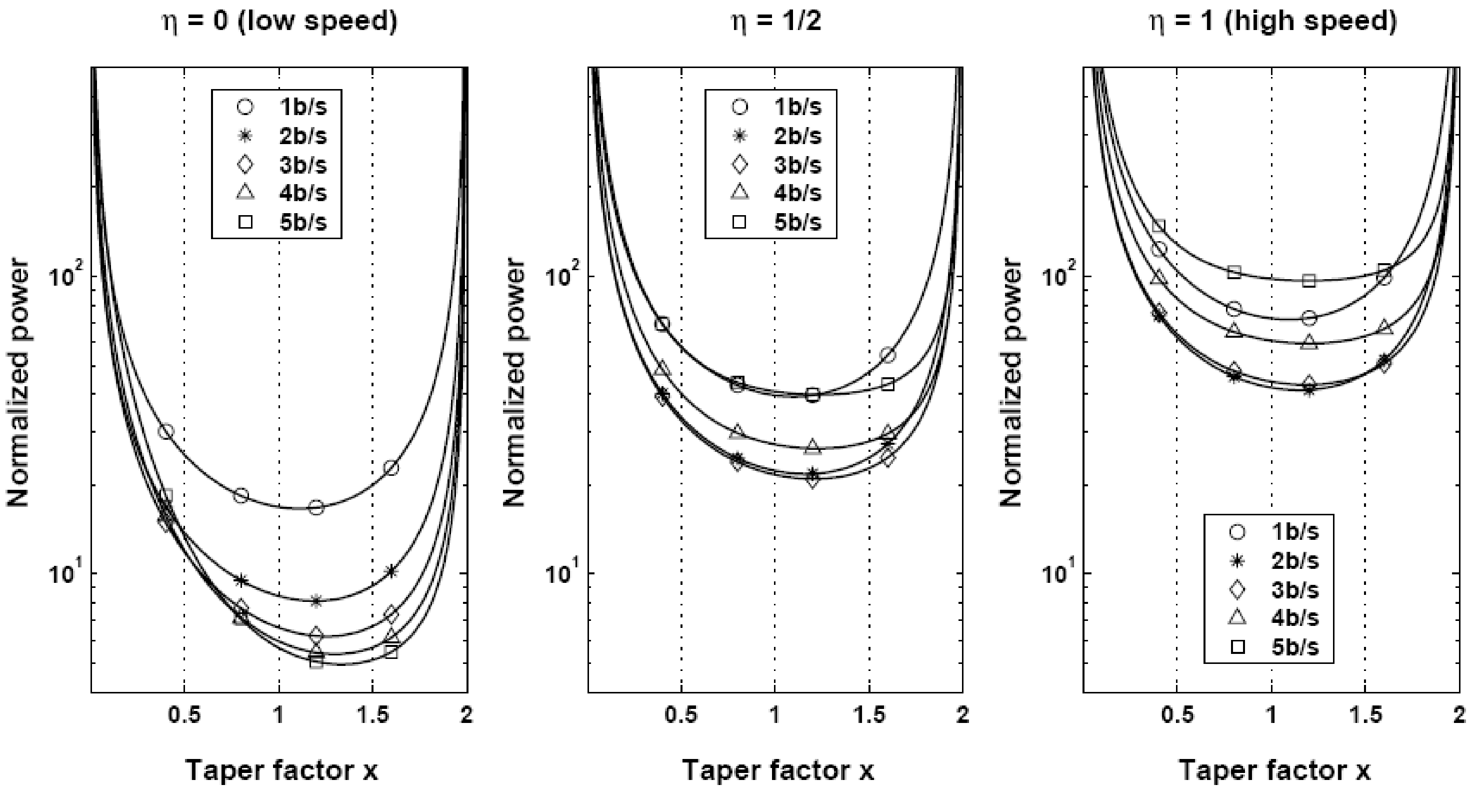
$$N_{tot} \times P_{tot} \propto \left[\frac{1}{C_1} + \frac{1}{4C_2} + \frac{1}{16C_3} + \cdots\right] \times [C_1 + C_2 + C_3 + \cdots]$$

$$\geq \left[1 + \frac{1}{2} + \frac{1}{4} + \cdots\right]^2$$

- Minimum is achieved when:
  - $C_1 = 2C_2 = 4C_3 = \ldots = 2^i C_{(i+1)}$
- The scaling factor is identical to the inter-stage gain

Shallow Optimum

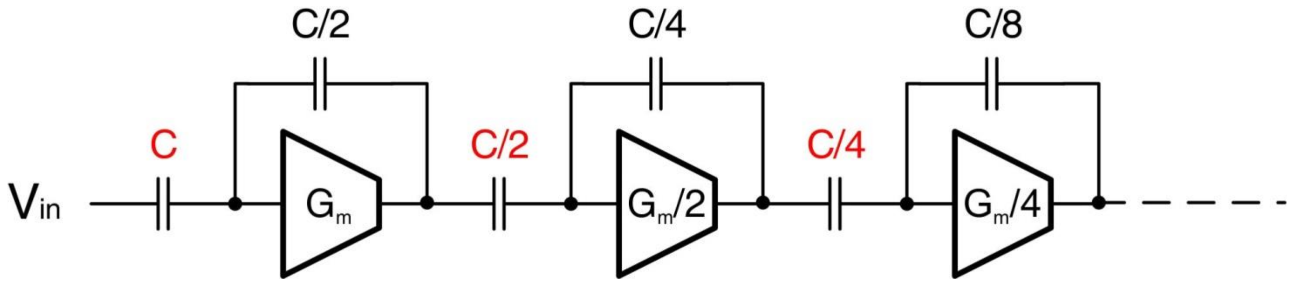
[Chiu 2004]



Capacitor scaling factor = 2^{Rx} Taper factor x=1 ⇒ scaling exactly by stage gain

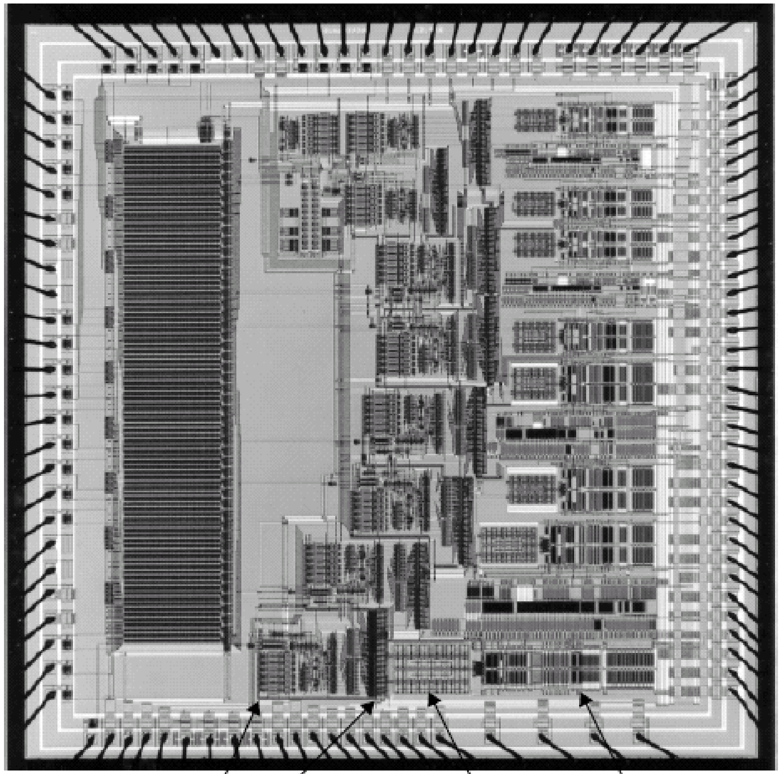
Practical Approach to Stage Scaling

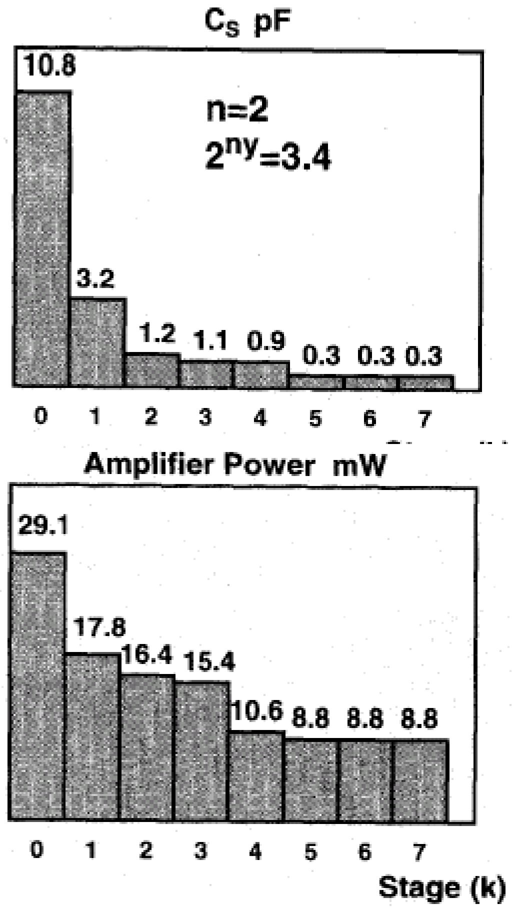
- Start by assuming caps are scaled precisely by stage gain
  - E.g. for 1-bit effective stages:



- Refine using circuit information
  - Use estimates of OTA power, parasitics, minimum feasible sampling capacitance etc.

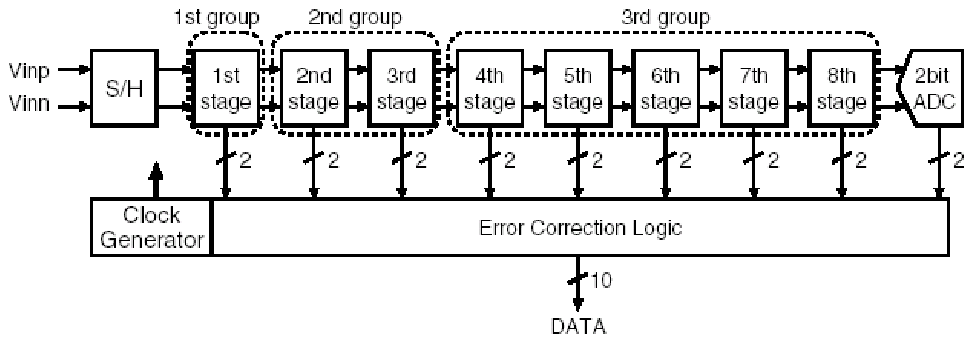
Stage Scaling Examples (1)



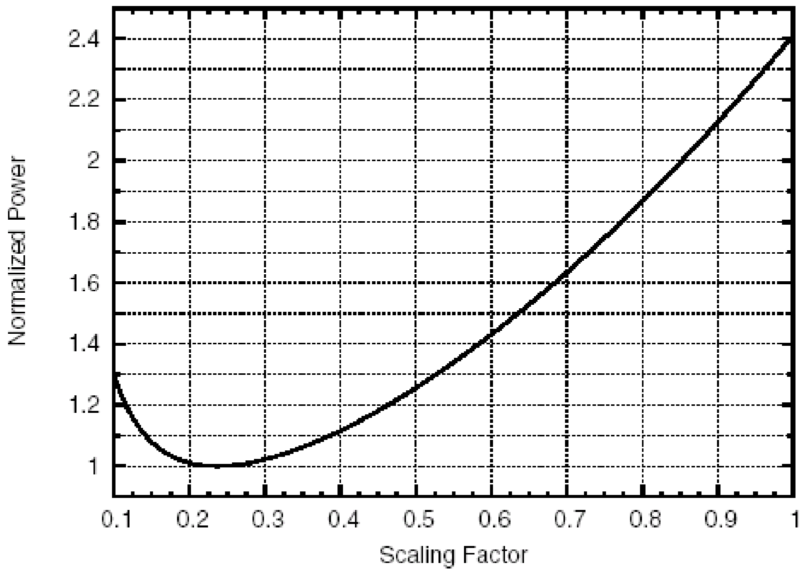


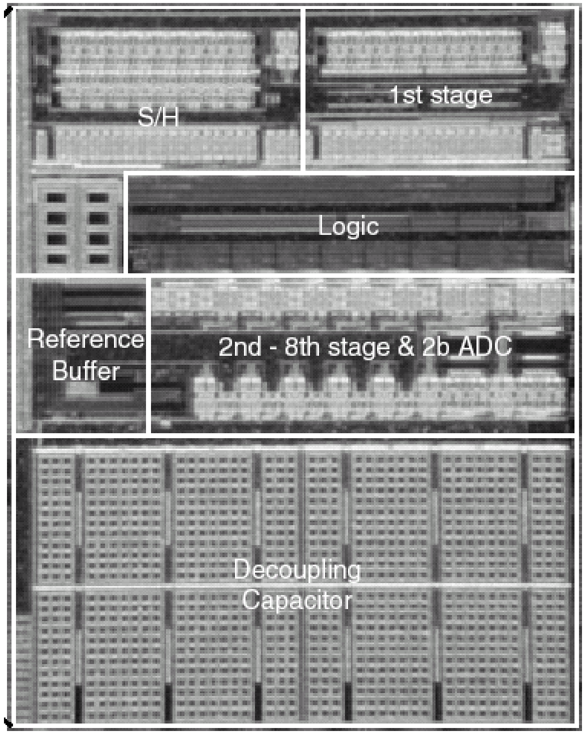
[Cline 1996]

Stage Scaling Examples (2)



[Ishii 2005]

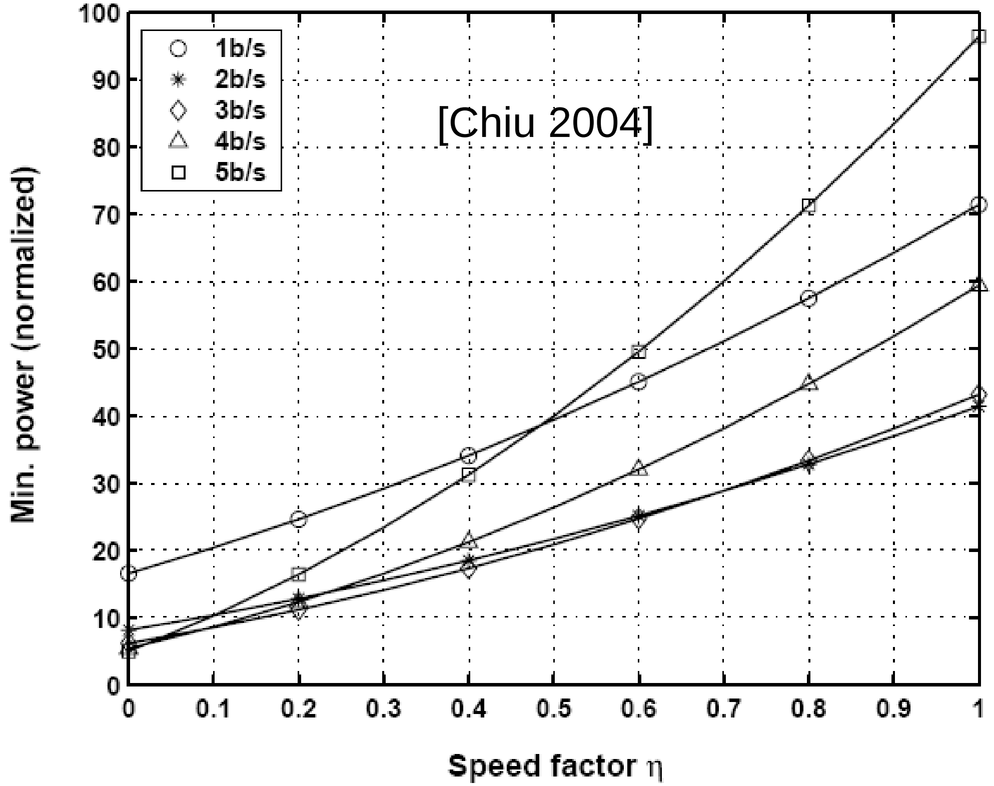




How Many Bits Per Stage?

- Small # bits per stage (e.g. $R = \log_2 G = 1$)
  - OTAs have small closed loop gain and large feedback factor
    - Suitable for high speed ☺
  - Need many stages ☹
- Large # bits per stage (e.g. $R = \log_2 G = 3$)
  - Fewer number of stages ☺
  - OTAs have small feedback factors and smaller bandwidths
    - Not suitable for high speed ☹
  - Need many comparators
    - Large loading from comparators ☹
- Qualitative conclusion
  - Use low per-stage resolution for very high speed designs
  - Try higher resolution stages when power efficiency is the most important constraint

Power Tradeoff is Fairly Flat!



η = parasitic cap at output/total sampling cap in each stage (junctions, wires, switches, …)

- ADC power varies by only ~2x across different stage resolutions!

Examples

Reference	[Yoshioka, 2007]	[Jeon, 2007]	[Loloee 2002]	[Bogner 2006]
Technology	90nm	90nm	0.18um	0.13um
Bits	10	10	12	14
Bits/Stage	1-1-1-1-1-1-1-3	2-2-2-4	1-1-1-1-1-1-1-1-1-1-2	3-3-2-2-4
SNDR [dB]	~56	~54	~65	~64
Speed [MS/s]	80	30	80	100
Power [mW]	13.3	4.7	260	224
mW/MS/s	0.17	0.16	3.25	2.24

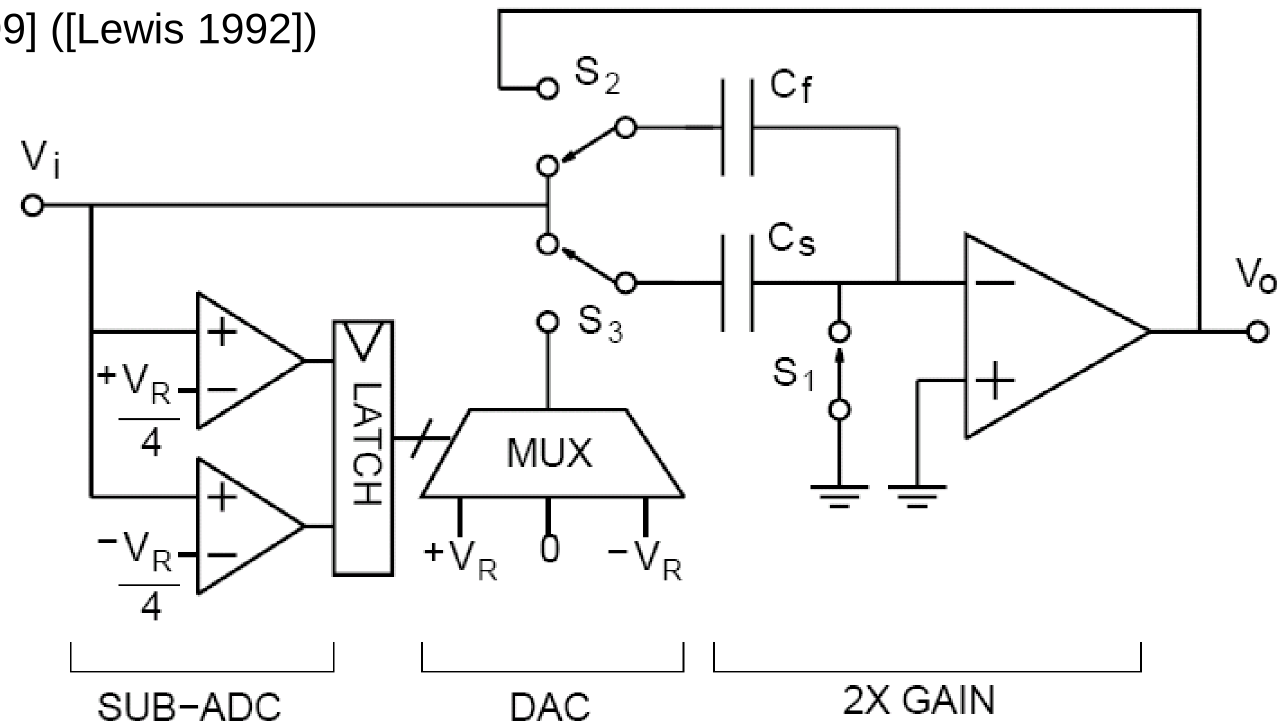
- Low power is possible for a wide range of architectures!

Brief Summary

- Choosing the "optimum" per-stage resolution and stage scaling scheme is a non-trivial task
  - But optima are shallow!
- Transistor level design and optimization is at least as important as architectural level optimization
- Next, look at circuit design details
  - Assume we're trying to build a 10-bit pipeline
    - 180nm process
    - Moderate to high-speed ~100MS/s
    - 1-bit effective/stage, using "1.5-bit" stage topology
    - Dedicated front-end SHA

1.5-Bit Stage Implementation

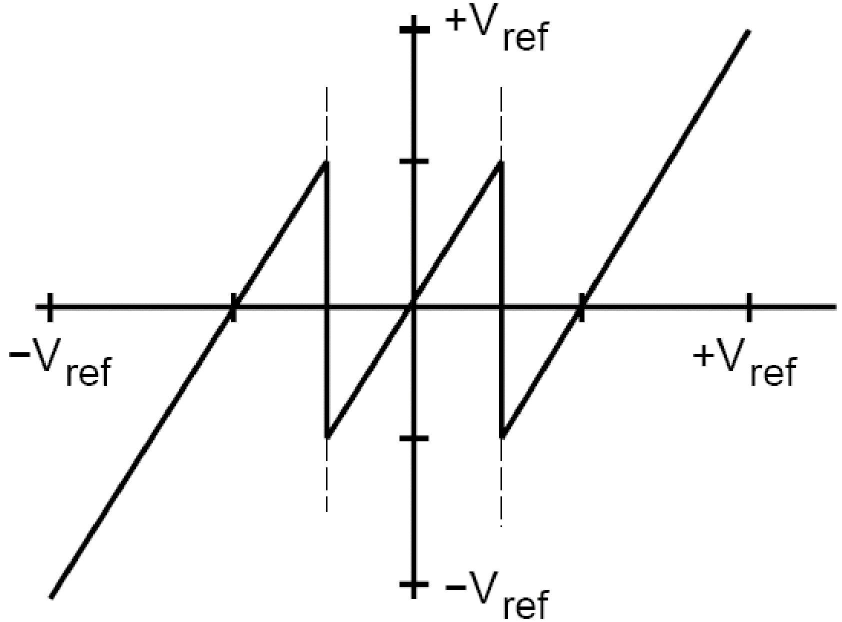
[Abo 1999] ([Lewis 1992])



- C_f is used as sampling cap during acquisition phase and feedback cap during charge redistribution phase
  - Enlarged feedback factor (1/3 → 1/2)

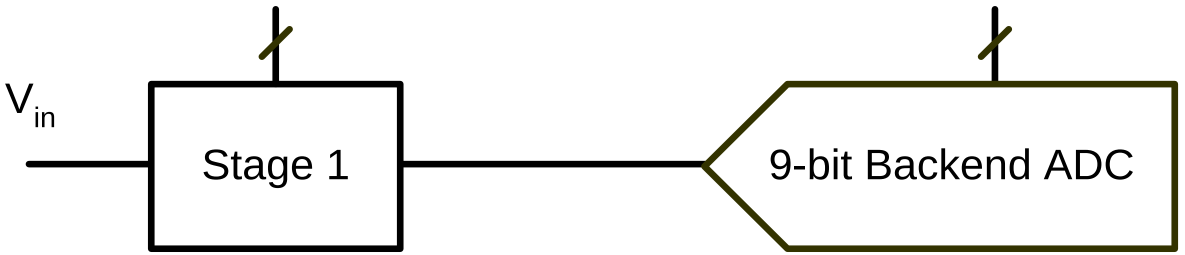
Residue Plot

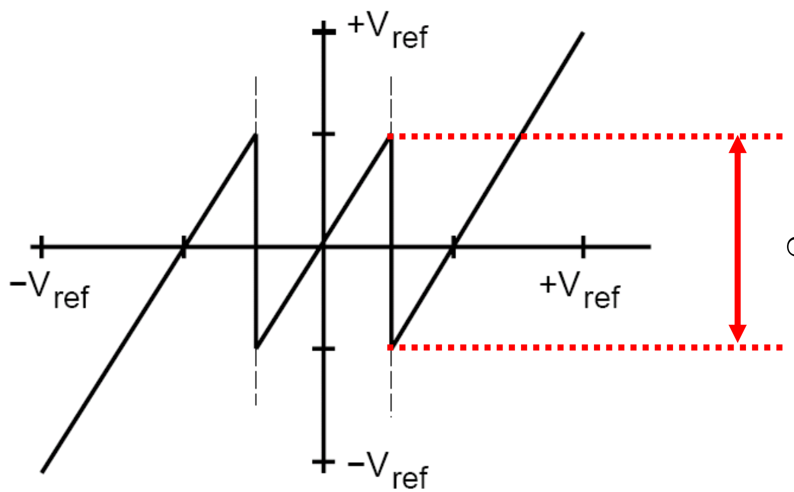
$$V_o = \begin{cases} \left(1 + \frac{C_s}{C_f}\right) V_i - \frac{C_s}{C_f} V_{ref} & \text{if } V_i > V_{ref}/4 \\ \left(1 + \frac{C_s}{C_f}\right) V_i & \text{if } -V_{ref}/4 \leq V_i \leq +V_{ref}/4 \\ \left(1 + \frac{C_s}{C_f}\right) V_i + \frac{C_s}{C_f} V_{ref} & \text{if } V_i < -V_{ref}/4 \end{cases}$$



[Abo 1999]

Stage 1 Matching Requirements





$$\propto \frac{C_s}{C_f} \cong 1 + \frac{\Delta C}{C}$$

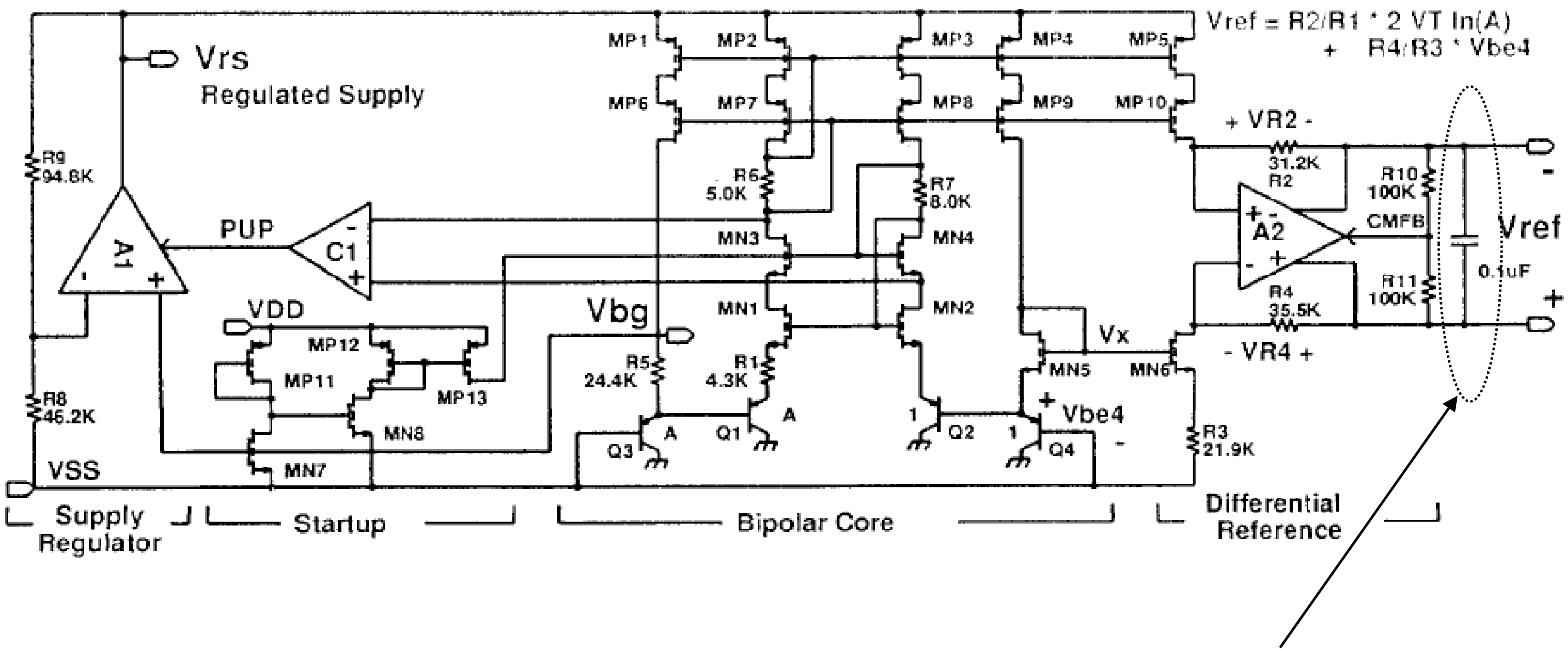
- Error in residue transition must be accurate to within a fraction of 9-bit backend LSB
  - Typically want $\Delta C/C$ ~0.1% or better

Capacitor Matching

- 0.1% "easily" achievable in current technologies
  - For MIM cap matching data see e.g. [Diaz 2003]
  - For MOM cap matching data see e.g. [Verma 2006]
  - Depend on capacitor size
- What if we needed much higher resolution than 10 bits?
  - Digital calibration
  - Multi-bit first stage
    - Each extra bit resolved in the first stage alleviates precision requirements on residue transition by 2x
      - Note that more capacitors are needed and thus capacitor area increases
      - For fixed first-stage total capacitance, does INL/DNL improve for going to multi-bit first stage?
      - For fixed-size unit capacitor, how does INL/DNL improve with increased number of bits in the first stage?
    - Multi-bit examples: [Singer 1996] [Kelly 2001] [Lee 2007]

Typical Reference Generator

[Todd Brooks 1994]



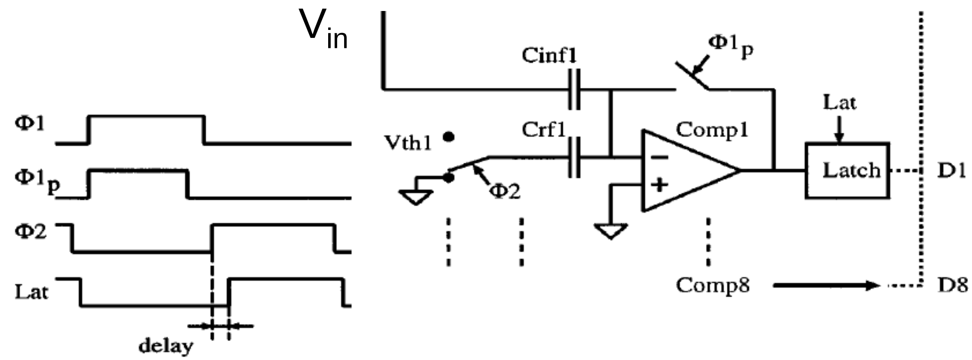
External decoupling caps provide dynamic currents

⇒ Low power reference buffer

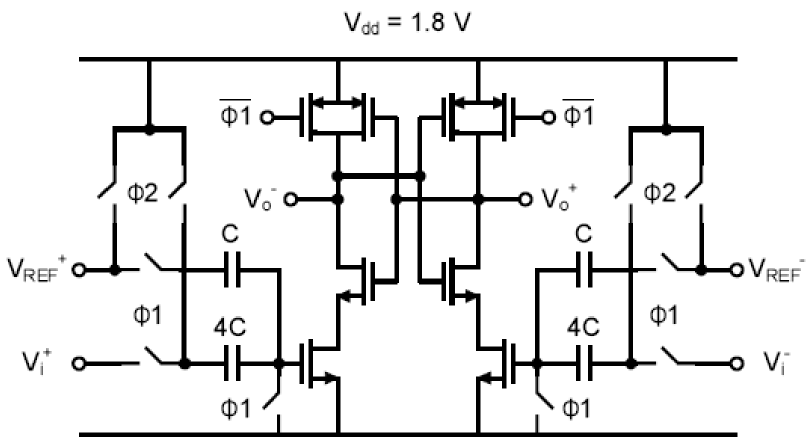
Comparators

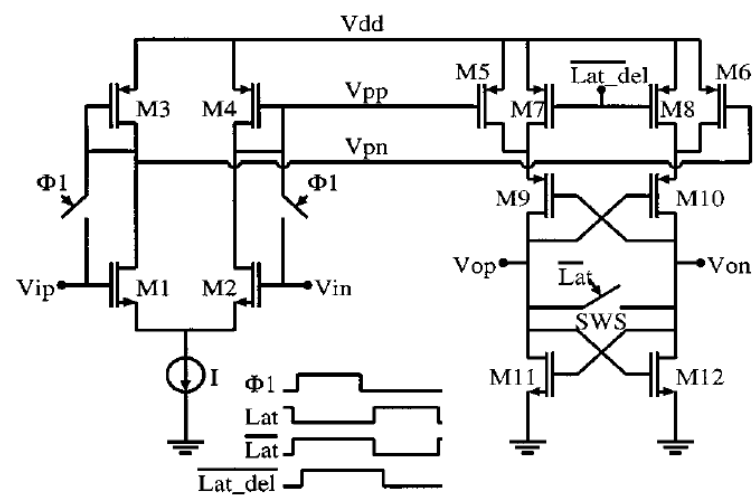
- Can tolerate large offsets and large noise with appropriate redundancy
- Consume negligible power in a good design
  - 50~100μW or less per comparator
- Lots of implementation options
  - Resistive/capacitive reference generation
  - Different pre-amp/latch topologies
  - …

Comparator Examples



[Chiu 2004]





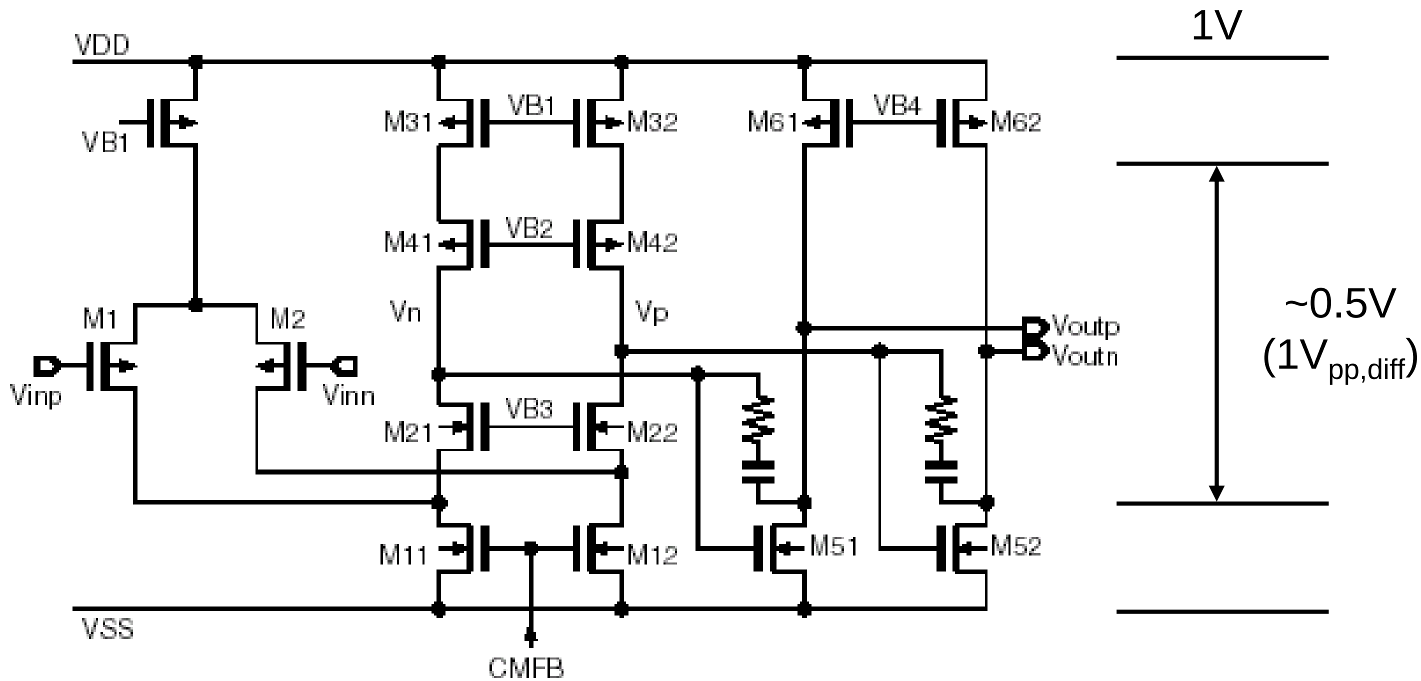
[Mehr 2000]

OTA Design Considerations

- Static amplifier error = 1/(Loop Gain)
  - For the first stage of 10-bit ADC, need loop gain > 60dB
- Dynamic settling error
  - Want outputs to settle to ~1/8 LSB accuracy within 1/2 clock cycle
- Thermal noise
  - Size capacitor sizes to satisfy noise requirement
- Start by picking an OTA topology that will deliver sufficient gain
  - Or think about ways to compensate finite gain error

Two-Stage Folded Cascode OTA

[Ishii 2005]



- Work down to V_{DD}=1V with reasonable output swing
- Gain ~ $(g_m r_o)^3$ ~ 10^3 = 60dB
- Use gain boosting to achieve larger gain

How Fast Can We Go? (1)

- Non-dominant pole in two-stage amplifier hard to move past $f_T/5$
- For 73° phase margin (optimum for fast settling), loop crossover frequency is 1/3 of non-dominant pole frequency
- Settling linearly to 0.1% precision takes 7 loop time constants; typically budget ~10 time constants
- Ideally, we'd have 1/2 clock cycle to settle linearly, but there is some time needed for slewing and non-overlap clock timing
  - Assume 60% of half cycle is available for linear settling
- In summary (here we have assume that the feedback factor is 1)

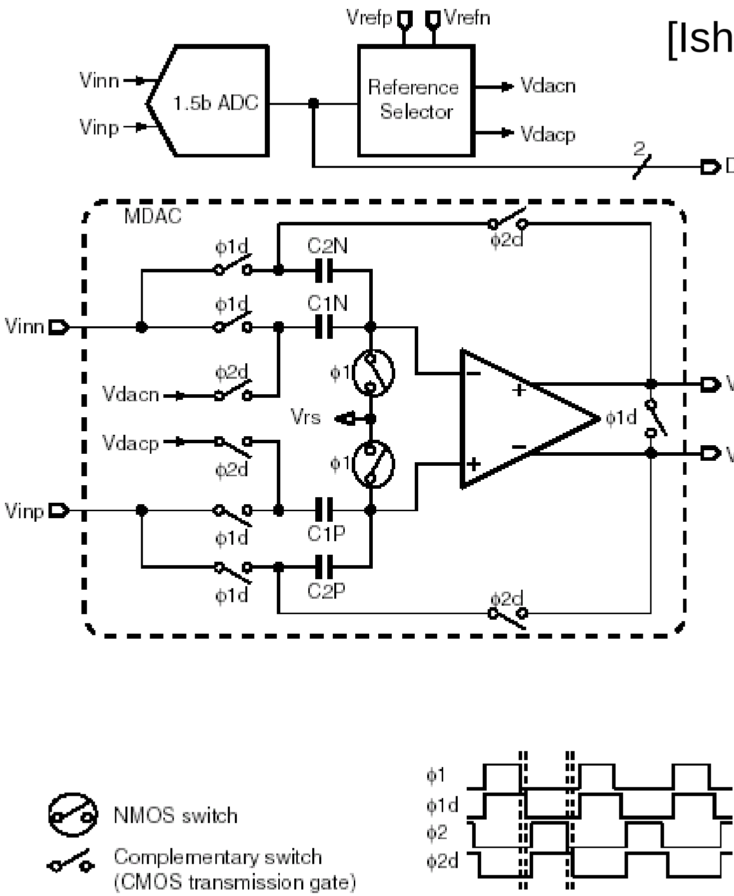
$$f_{CLK,max} = \frac{f_T}{5}\frac{1}{3}\cdot 2\pi \cdot \frac{1}{10} 0.5 \cdot 0.6 = \frac{f_T}{80}$$

How Fast Can We Go? (2)

Technology	NMOS f_T (at moderate V_{GS}-V_t ~150mV)	$f_{CLK,max} = f_T/80$
0.35um	10GHz	125 MHz
0.18um	30GHz	375 MHz
90nm	90GHz	1.125GHz

- Sampling speeds of 200-300MHz are "easily" achievable in today's technologies
  - f_T is no longer a showstopper
  - Speed ultimately constrained by power and clock jitter

Switches



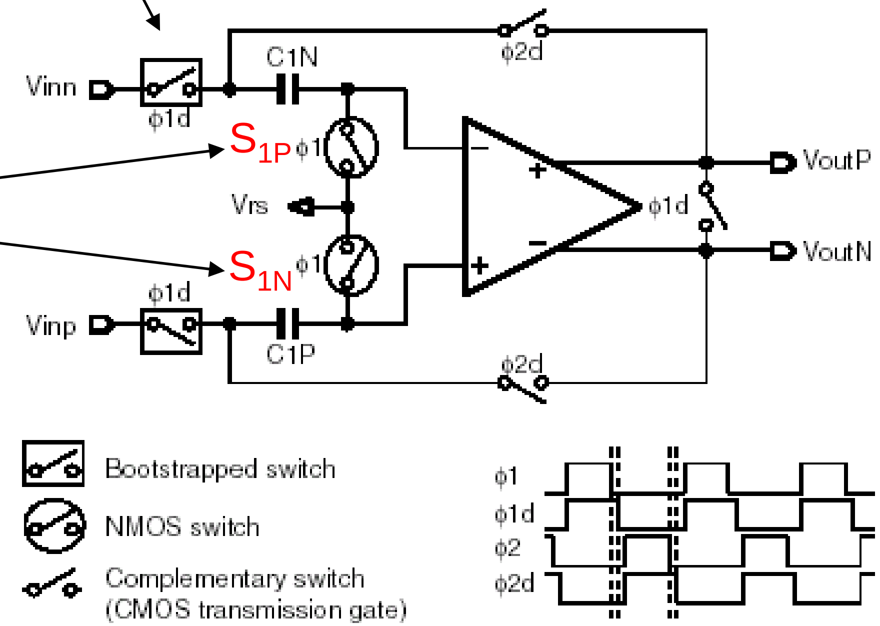
[Ishii 2005]

- Make switch RC ~ 10 times faster than OTA
  - Avoid speed degradation
  - Minimize switch noise contribution
    - See e.g. [Schreier 2005]
  - Avoid stability issues due to poles in feedback network
- Three choices for switches
  - Single N or P device
  - Transmission gate
  - Bootstrapped NMOS
    - For high swing nodes that require constant R_{on}

Front-End SHA

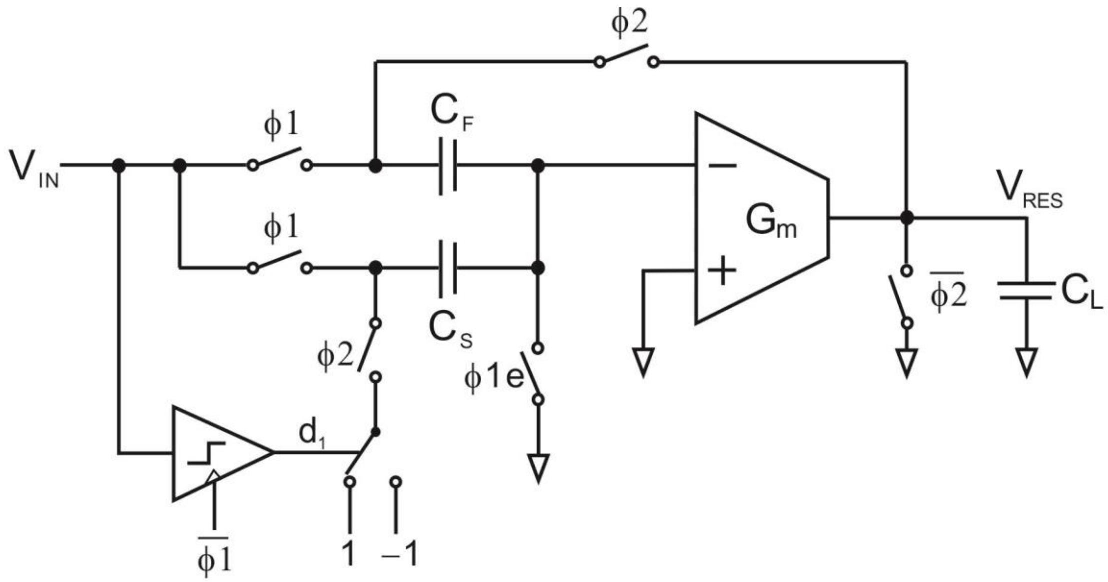
Need constant R_{ON} here to minimize signal dependent charge injection from S_{1N}, S_{1P}

Minimize Jitter!

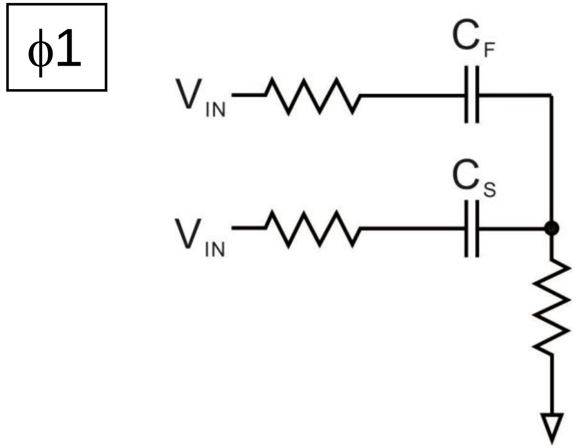


[Ishii 2005]

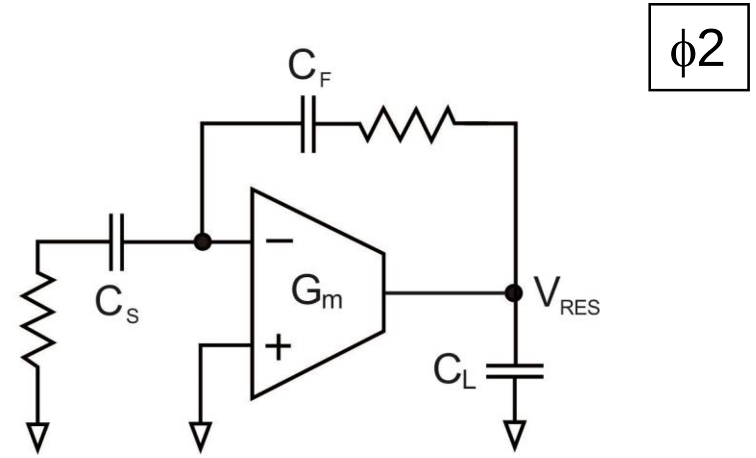
Settling



$\phi 2$
$\phi 1$
C_F
V_{IN}
$\phi 1$
G_m
V_{RES}
C_S
$\overline{\phi 2}$
C_L
$\phi 2$
$\phi 1e$
d_1
$\overline{\phi 1}$
1
−1

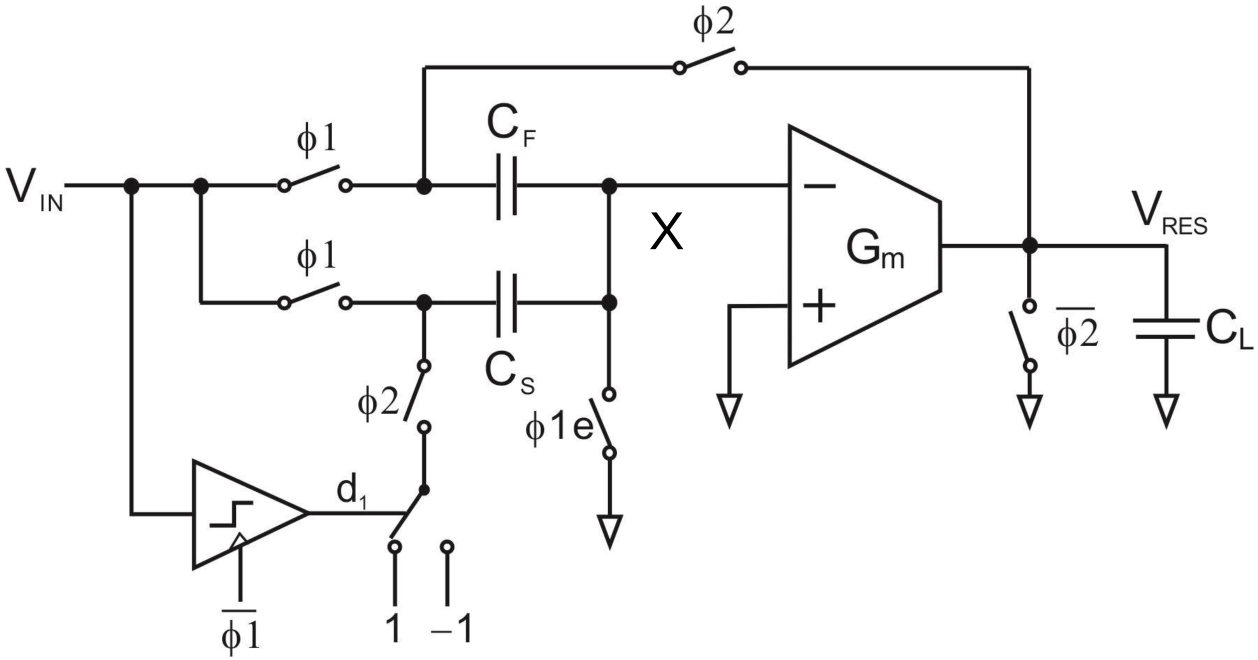


$\phi 1$
C_F
V_{IN}
C_S
V_{IN}



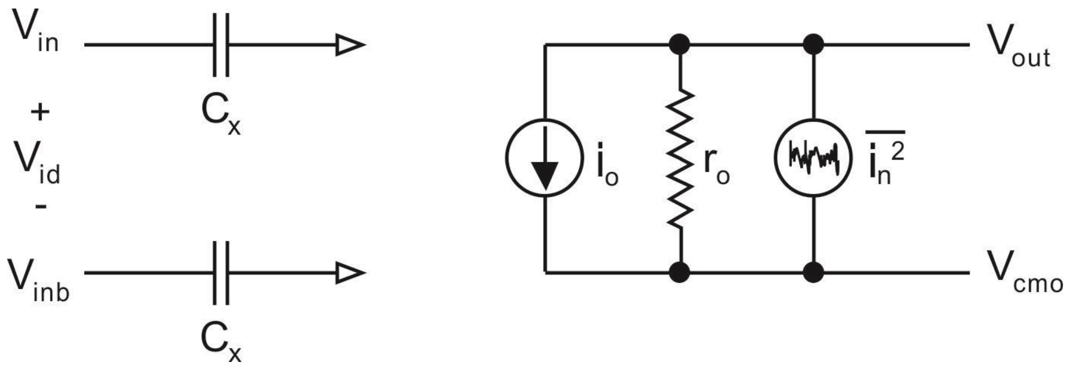
$\phi 2$
C_F
C_S
G_m
V_{RES}
C_L

Example



Parameters: f_s = 100MHz, R_{on} = 100Ω, C_S = C_F = 1pF, C_L = 1pF

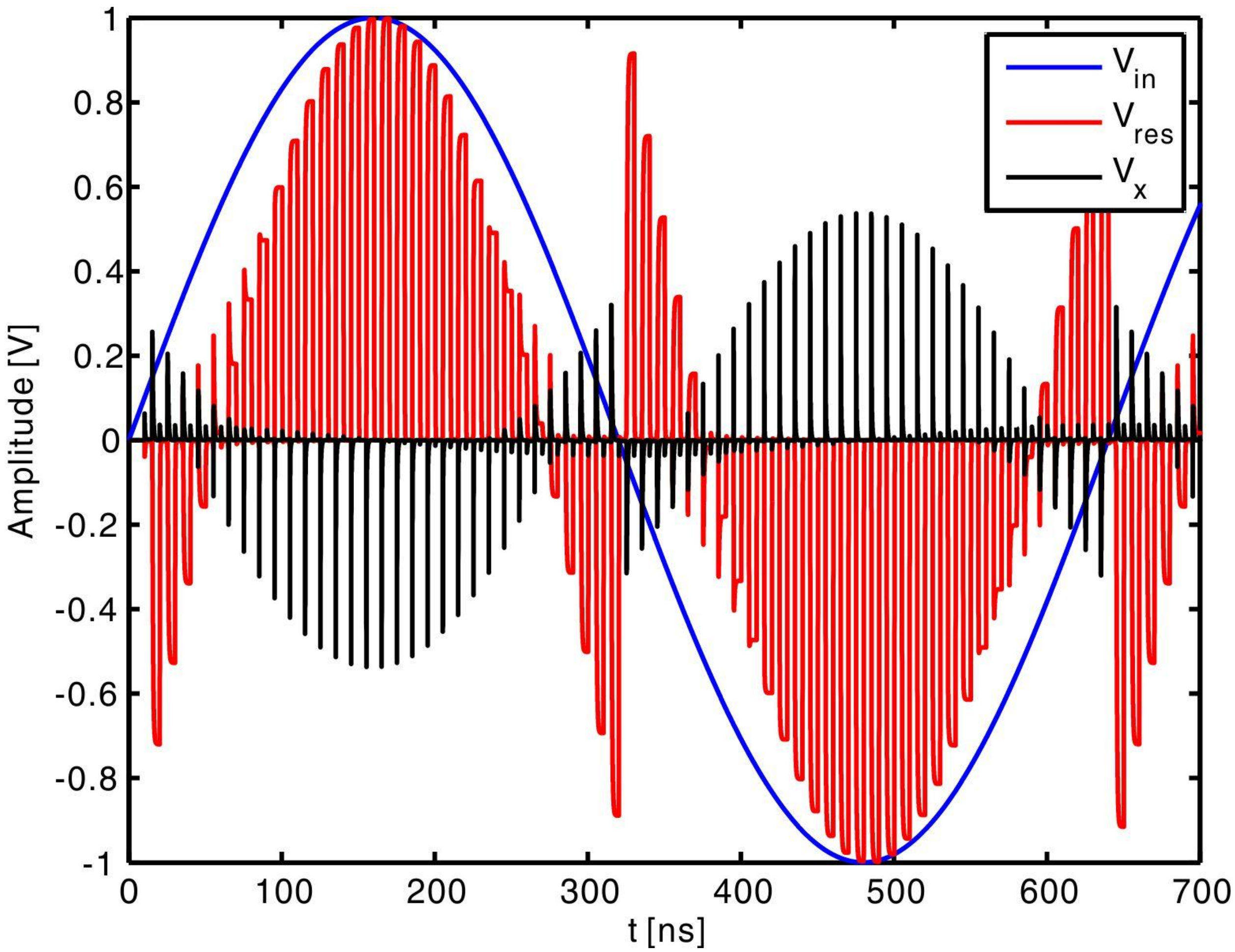
OTA Simulation Model



$$C_X = \frac{g_m}{2\pi f_T} \qquad \frac{\overline{I_n^2}}{\Delta f} = 8kTg_m \qquad r_o = \frac{A_V}{g_m}$$

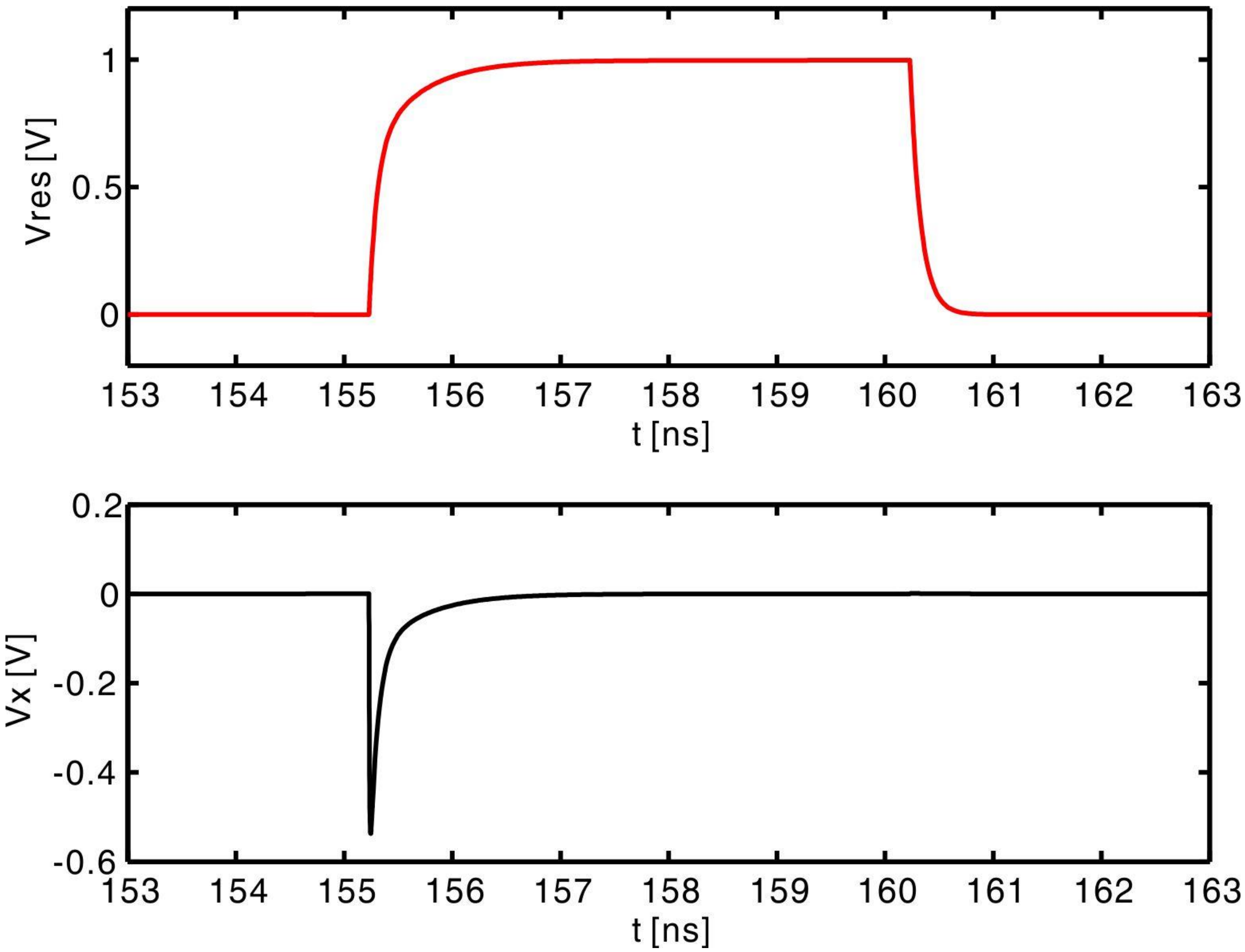
Parameters: g_m=5mS, A_V=1e3, g_m/I_D = 10S/A, f_T=20GHz

1-bit Pipeline Stage Behavior



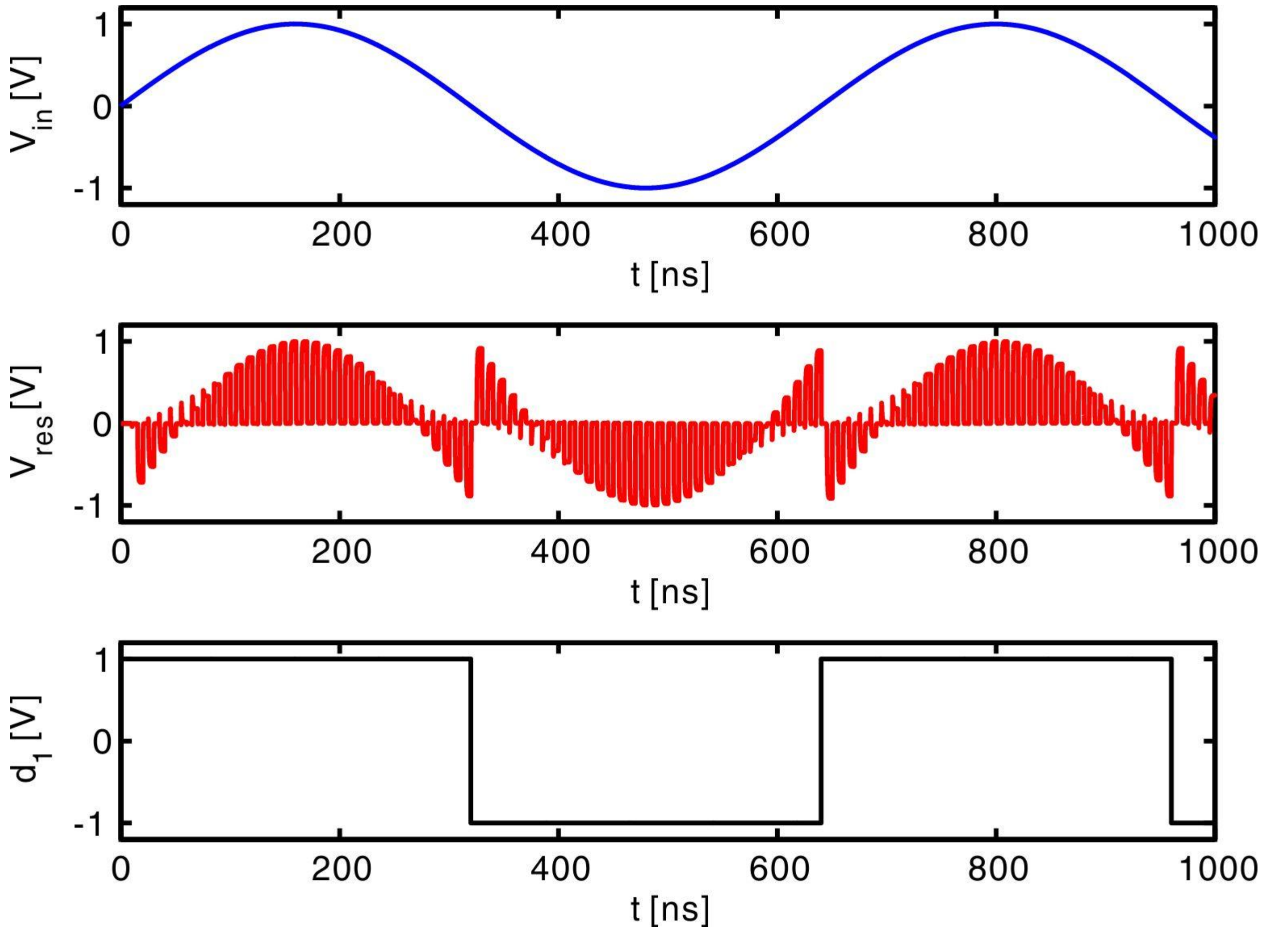
Jump in V_{res} when V_{in} crosses 0

Zoom in around V_{in} = 1V



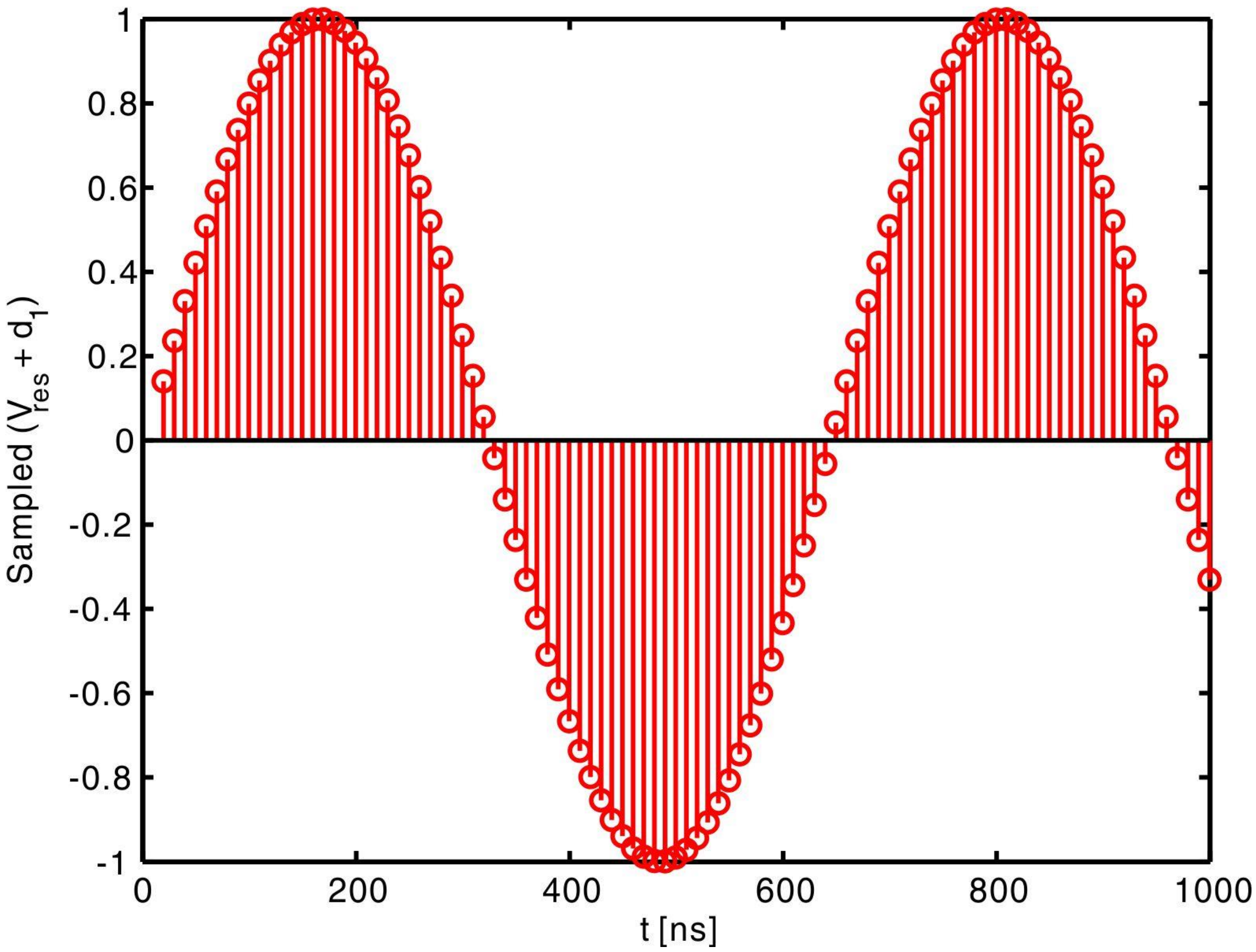
Linear settling in ϕ2

Pipelined Stage Output

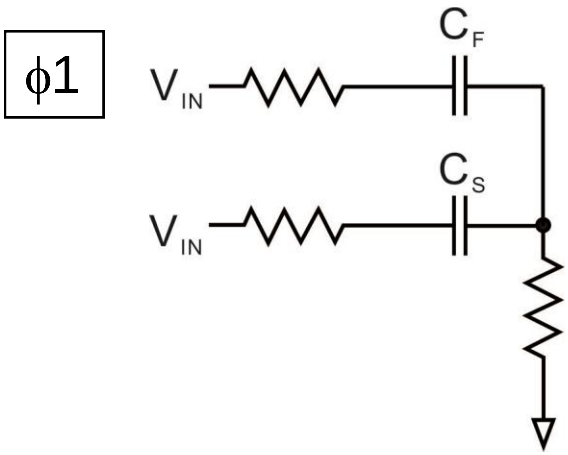


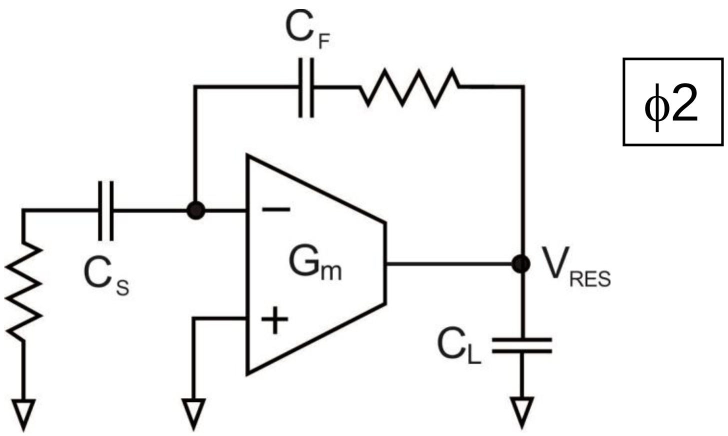
Question: how to test linearity for a pipelined stage?

Sampled $(V_{res} + d_1)/2$



Noise Analysis





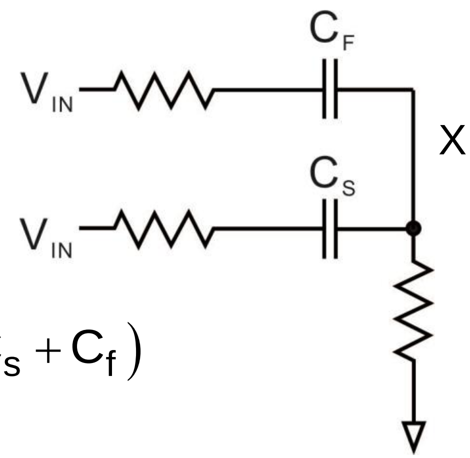
Noise due to switches

Noise due to OTA and switches

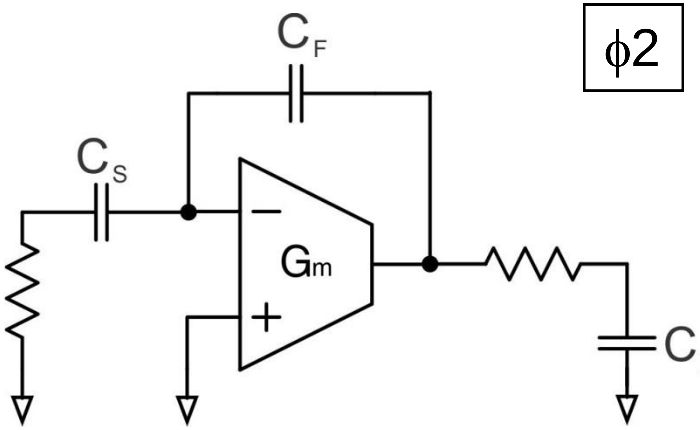
Identical to that of the closed-loop track-and-hold

Noise Analysis

$\boxed{\phi 1}$



$$\overline{q_x^2} = kT\left(C_s + C_f\right)$$



$\boxed{\phi 2}$

$$\overline{v_{o,1}^2} = \frac{\overline{q_x^2}}{C_f^2} = kT\left(\frac{C_s + C_f}{C_f^2}\right) = \frac{kT}{C_f}\left(1 + \frac{C_s}{C_f}\right)$$

$$\overline{v_{o,2}^2} \cong \alpha\gamma \frac{1}{\beta}\frac{kT}{C_{Ltot}}$$

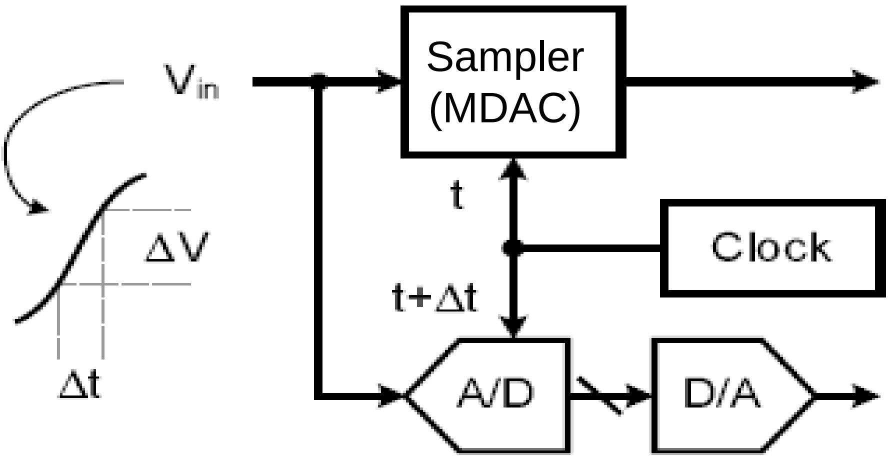
$$\boxed{\overline{v_{o,tot}^2} = \frac{kT}{C_f}\left(1 + \frac{C_s}{C_f}\right) + \alpha\gamma\frac{1}{\beta}\frac{kT}{C_{Ltot}}}$$

Noise Budgeting

- Total input referred noise budget, assuming $V_{FS,diff}$=1V
  - $N_{thermal}=N_{quant}=LSB^2/12=(1V/2^{10})^2/12=(280\mu Vrms)^2$
- First-order partitioning of input referred noise:
  - SHA $\rightarrow$ 1/3
  - Stage 1 $\rightarrow$ 1/3
  - All remaining stages $\rightarrow$ 1/3

SHA-Less Architectures (1)

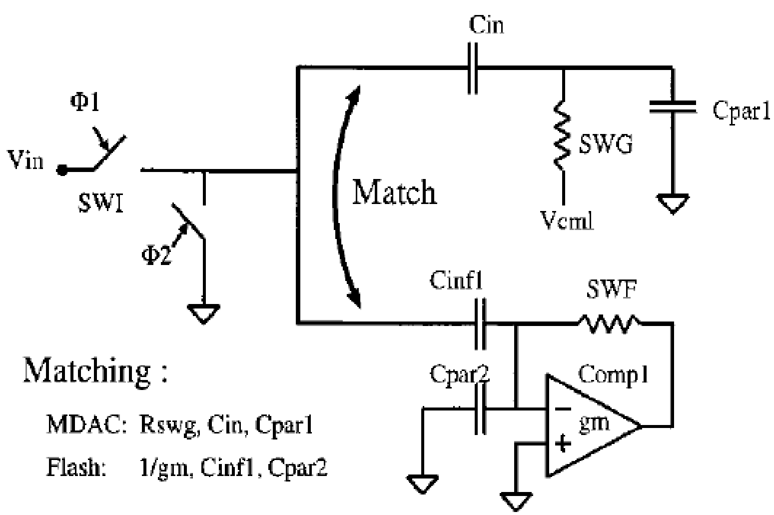
- SHA can burn up to 1/3 of the total ADC power
- For low power design, we want to removing front-end SHA
  - It causes acquisition timing mismatch between 1st stage MDAC and flash ADC



[Chiu 2004]

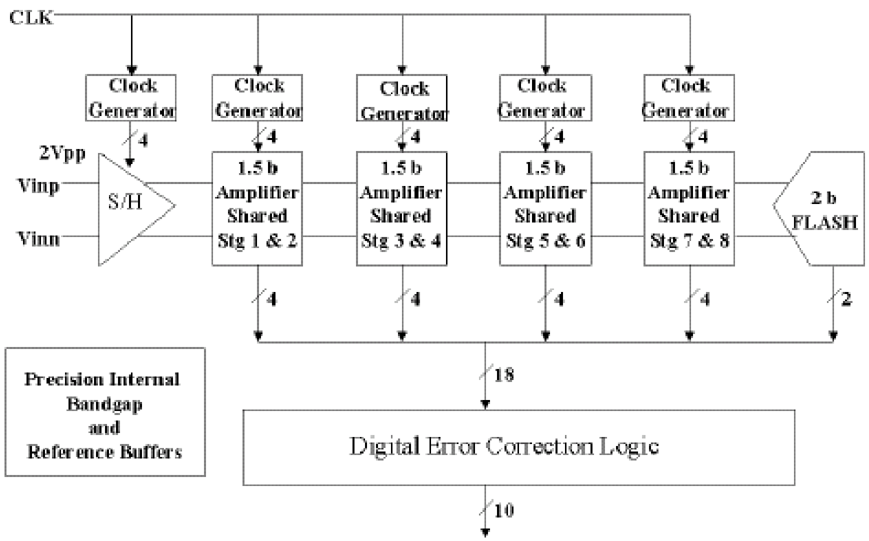
SHA-Less Architectures (2)

- Solutions
  - Use low-bit 1st stage with large redundancy
    - This can help absorb fairly large skew errors
  - Try to match sampling sub-ADC/MDAC networks
    - Bandwidth and clock timing

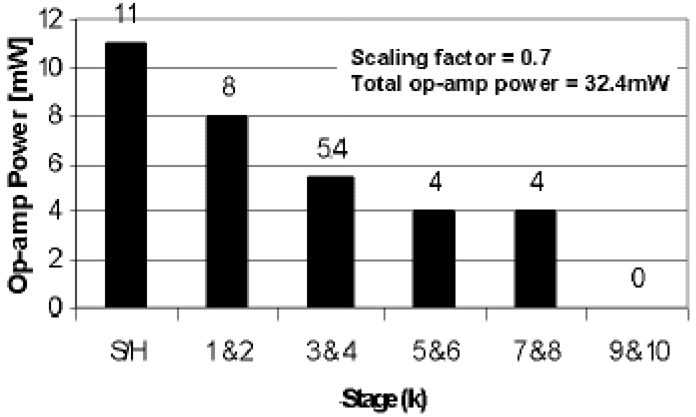


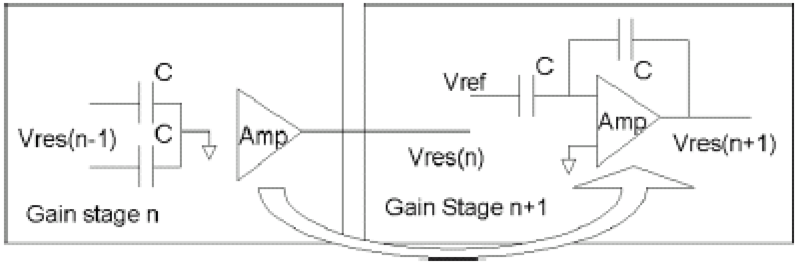
[Mehr 2000]

Amplifier Sharing (1)



[Min 2003]

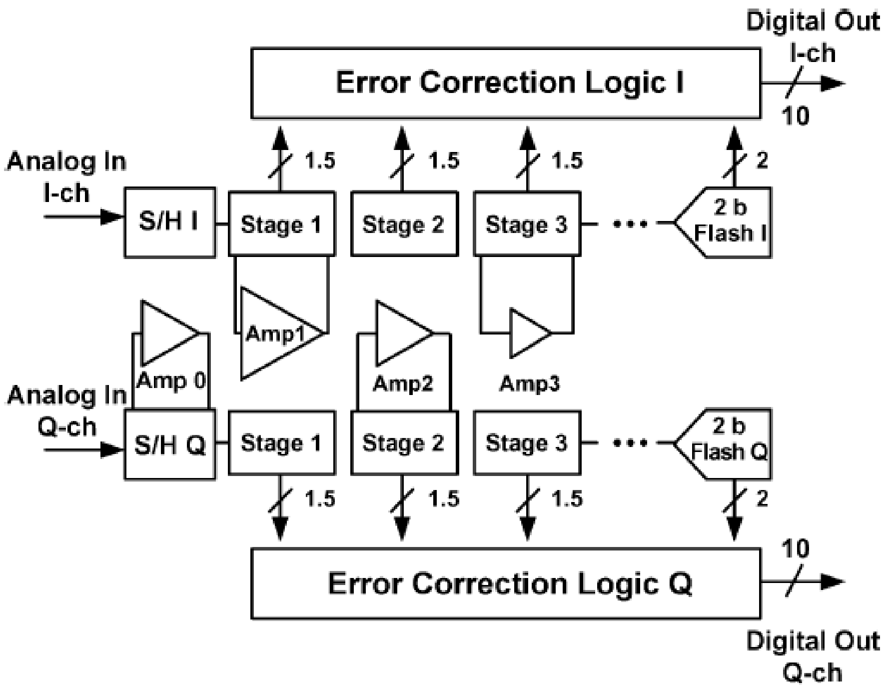




- Limited power savings because amplifiers have different specs
- We can turn off the OTA when not in use; however, be careful about memory effects

Amplifier Sharing (2)

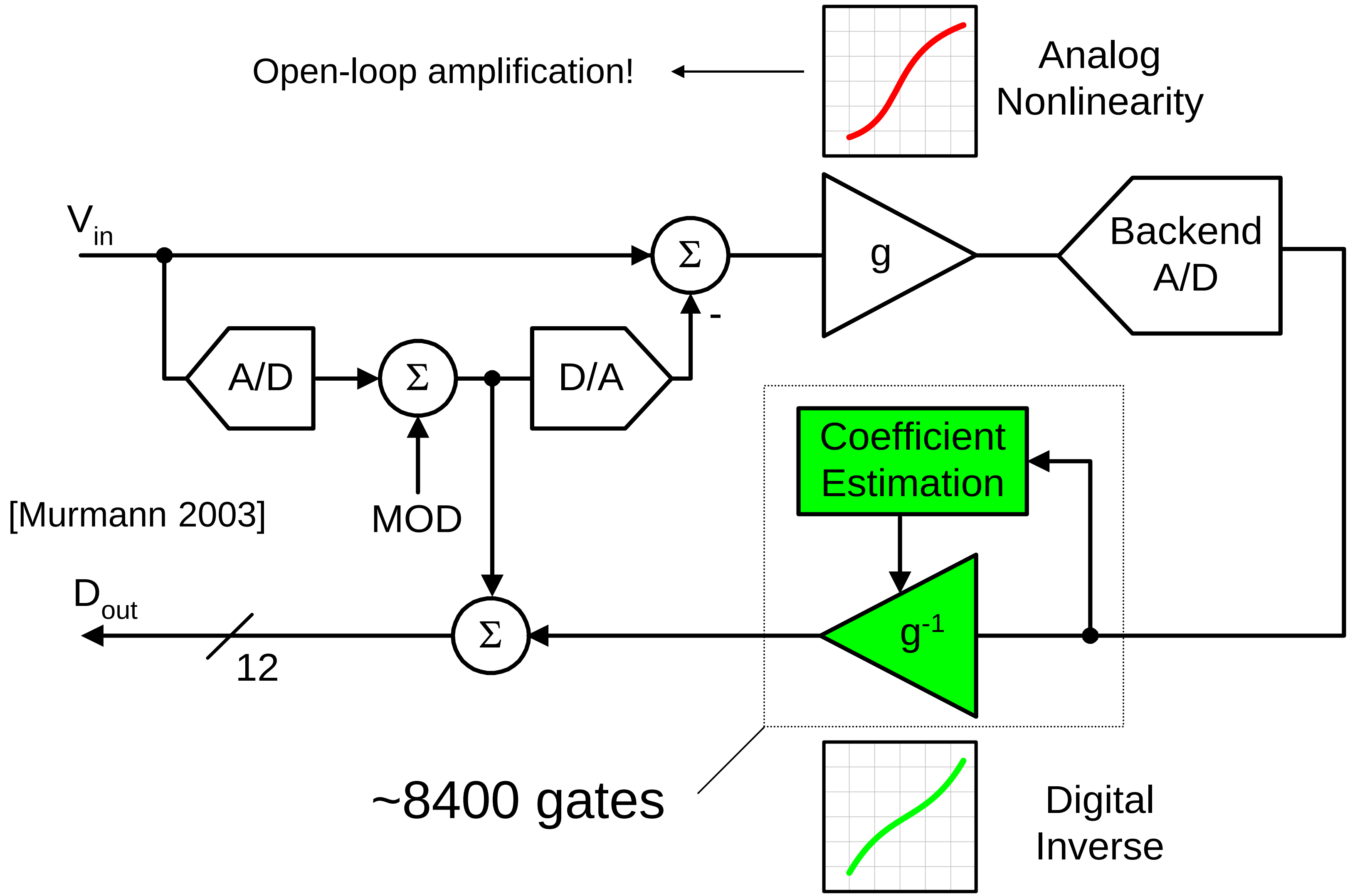
[Kurose 2006]



Technology	90-nm 1-P 7-M CMOS
Supply Voltage	1.2 V
Resolution	10 bit
Sampling Rate	200 MSPS
Full Scale	$0.8V_{p\text{-}p}$
DNL	+0.66/-0.61 LSB
INL	+0.90/-1.00 LSB
SFDR	66.5 dB
SNR	57.4 dB@Fin=9.9 MHz 55.6 dB@Fin=89.9 MHz
SNDR	54.4 dB@Fin=9.9 MHz 53.6 dB@Fin=89.9 MHz
ENOB	8.7 bit@Fin=9.9 MHz 8.6 bit@Fin=89.9 MHz
I/Q Isolation	>59 dB
Area	1.8 mm × 1.4 mm (2 ch)
Power	54.6 mW/ch

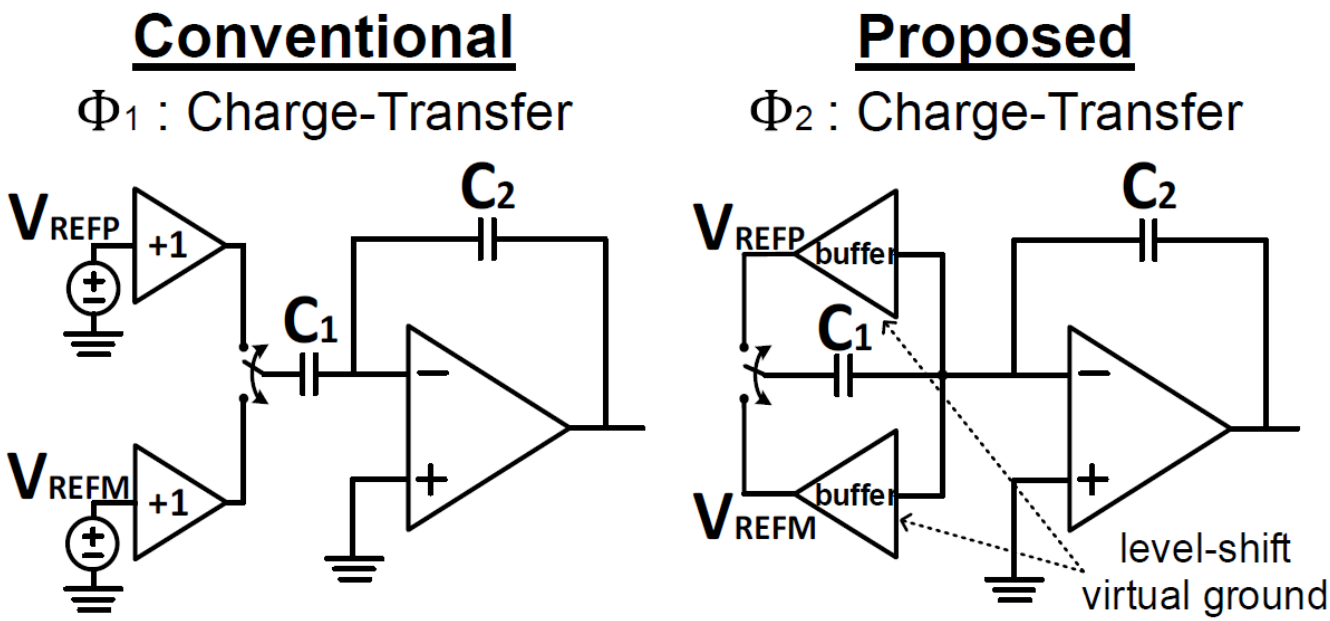
- Sharing of amplifiers is most efficiently done in a pair of converters that process I/Q signals

Research Topics



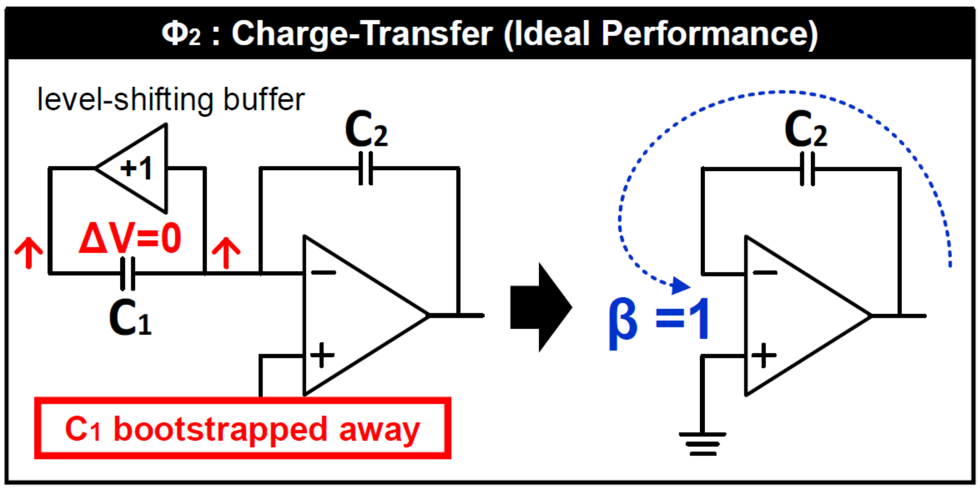
Research Topics

[H. Boo, ISSCC, 2015]



Research Topics

[H. Boo, ISSCC, 2015]



	Conventional	Proposed
G_s signal gain	$1+\frac{C_1}{C_2}$	$1+\frac{C_1}{C_2}$
β feedback factor	$\frac{C_2}{C_1+C_2}$	1
Closed-loop bandwidth	$\frac{f_u}{G_s}$	f_u
Input-referred noise density	$\overline{V_n^2}$ **	$\frac{\overline{V_n^2}}{G_s^2}$ ***
Charge-transfer error	$\frac{G_s}{A^*}$	$\frac{1}{A}$

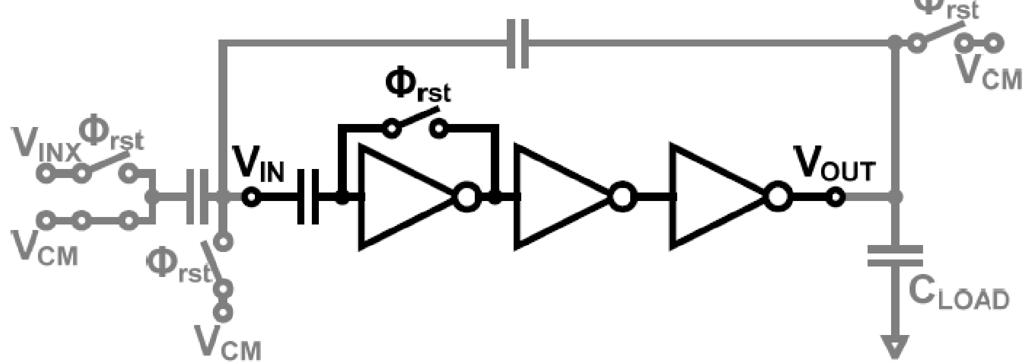
* op-amp DC gain

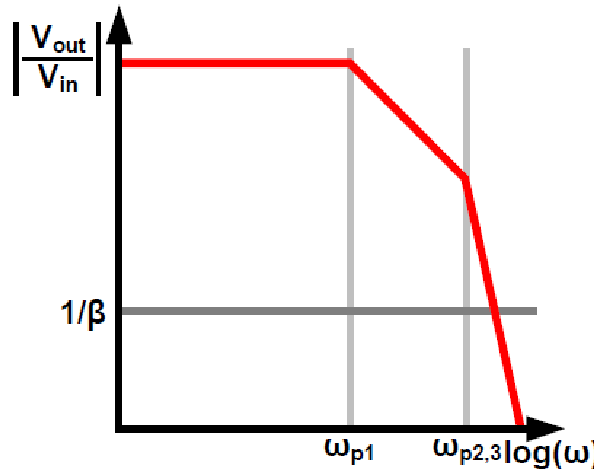
** op-amp noise density, noise bandwidth = f_u/G_s

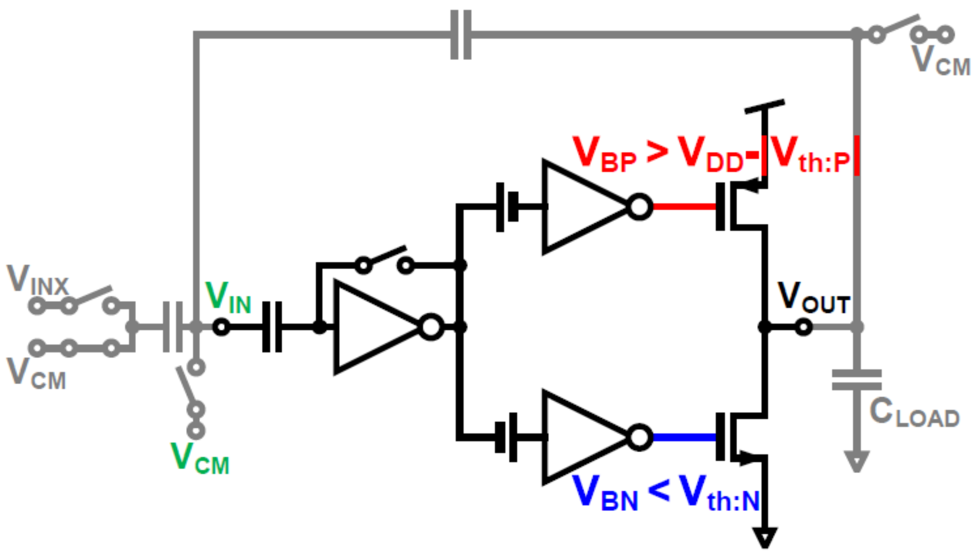
*** op-amp noise density, noise bandwidth = f_u

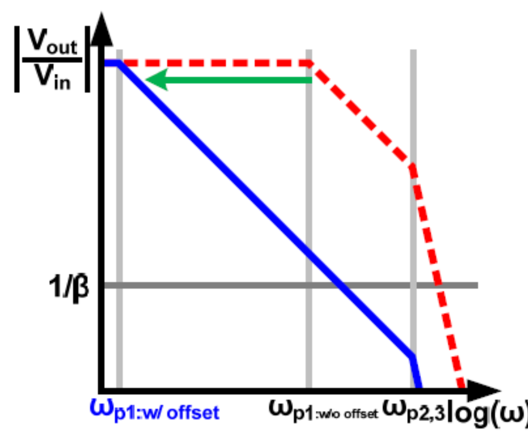
Research Topics

[Y. Lim, ISSCC, 2015]

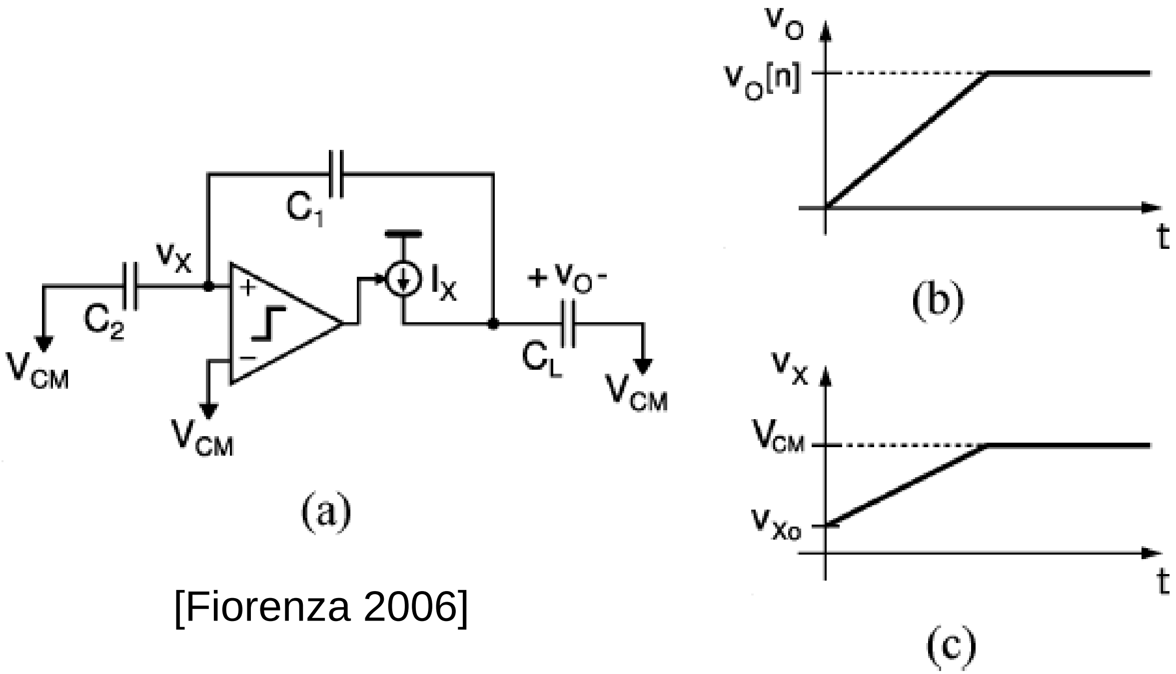








Research Topics

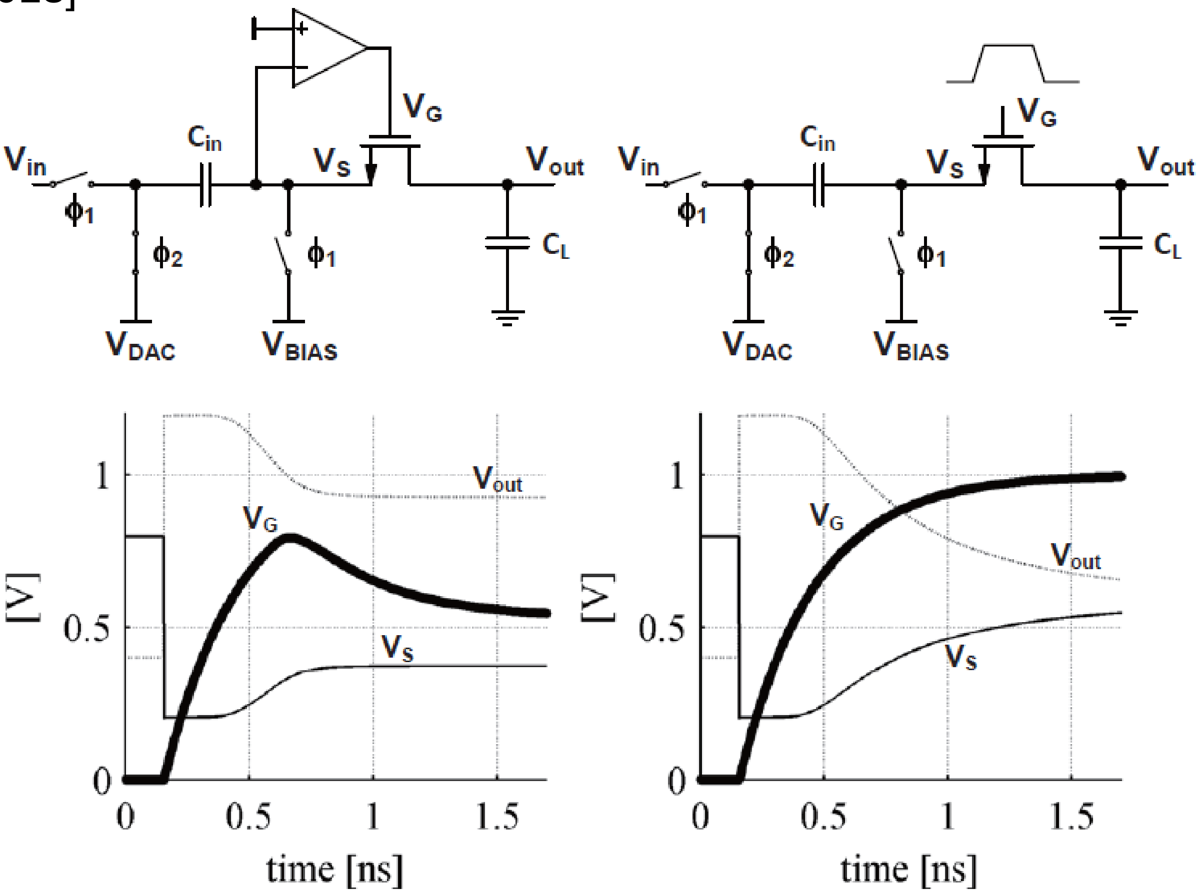


[Fiorenza 2006]

- Comparator-based switched-capacitor circuits
- Output slews to final value
  - Most efficient way to transfer charge

Research Topics

[N. Dolev, VLSI, 2013]

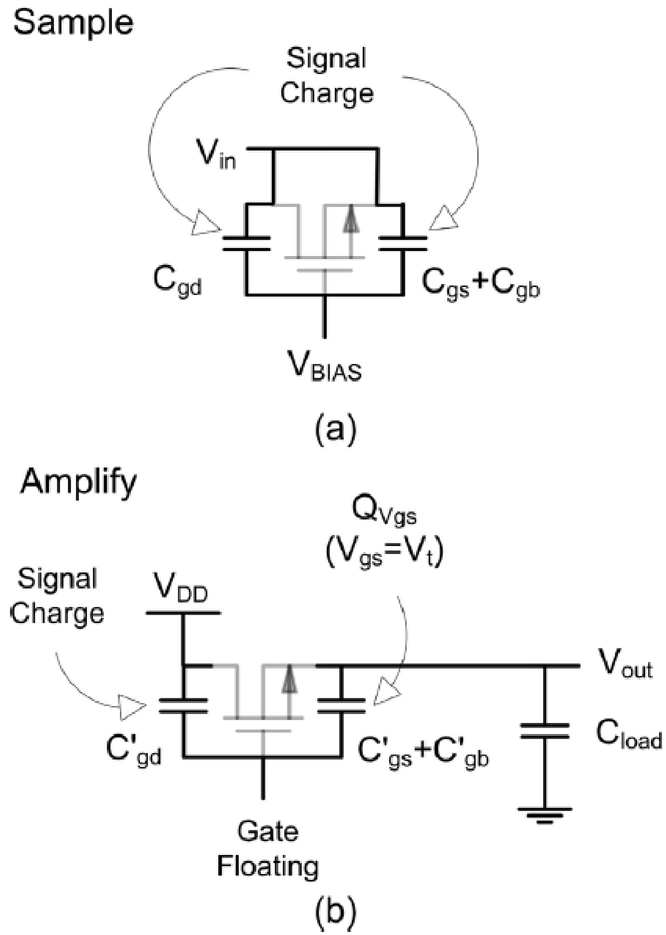


(a) Boosted Bucket Brigade Circuit

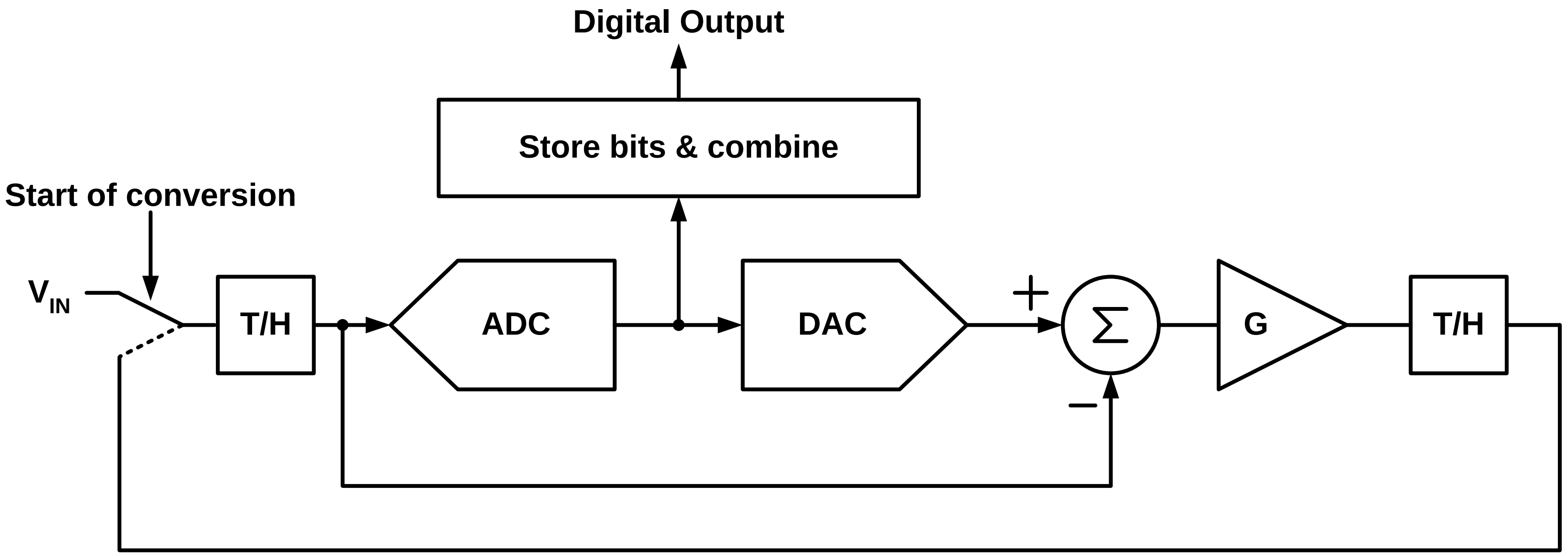
(b) Pulsed Bucket Brigade Circuit

Research Topics

[J. Hu, JSSC, 2009]



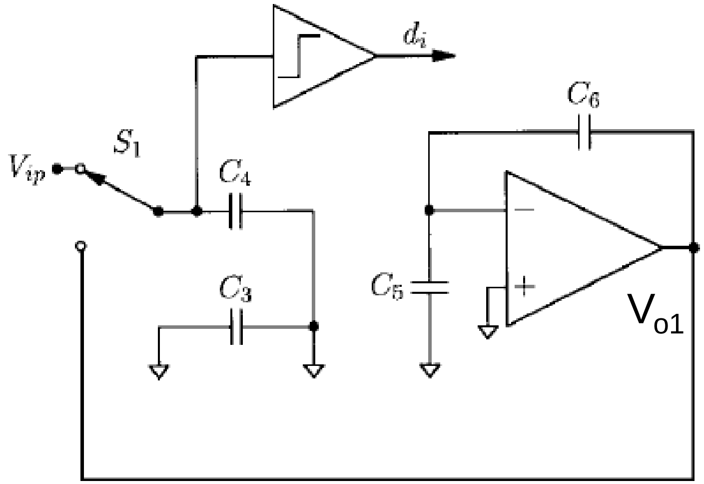
Cyclic ADC



- Essentially same as pipeline, but a single stage is used in a cyclic fashion for all operations
- Need many clock cycles per conversion

Implementation Example

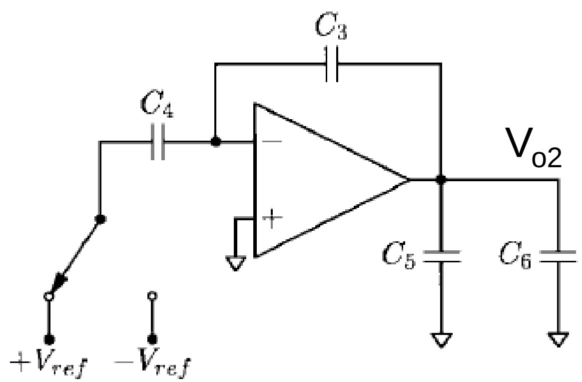
$\phi 1$



(a)

$$V_{o1} = \left(1 + \frac{C_5}{C_6}\right) V_{o2}$$

$\phi 2$



[Erdogan et al. JSSC 12/99]

$$V_{o2} = \left(V_{ip} \pm V_{ref}\right) \frac{C_4}{C_3}$$

Discussion

- Advantages
  - Area efficient
    - Only one or two switched capacitor stages plus comparator
  - Easy to calibrate
    - Only one stage
- Disadvantages
  - Slow
    - Need many clock cycles for a single conversion
  - Sub-optimal power efficiency
    - Cannot scale stages like in a pipeline ADC
    - Noise and accuracy requirements decrease from MSB to LSB cycle, but invested circuit energy per cycle is (usually) constant